

Connectivity Logic of Dendritic Spines in Cortex: Increased Inputs and Ensemble Formation

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ABSTRACT

Dendritic spines, small protrusions covering the dendrites of most neurons, are fundamental elements of synaptic connectivity, yet their network-level organization remains poorly understood. Here we leverage the large-scale MICrONS volumetric electron microscopy dataset of mouse primary visual cortex to explore the connectivity logic of dendritic spines across multiple spatial scales. Our analysis provides structural support for the “connectivity and diversity” hypothesis, showing that, for both excitatory and inhibitory neurons, dendritic spine density correlates with the number and the diversity of presynaptic partners. We further show that the preference for excitatory axons to form synapses on spines increases with distance from the soma. In addition, we find that spines serve as postsynaptic targets for high-output neurons (“out hubs”). We finally uncover an input/output correlation, in which neurons that share their input also preferentially target spines through their axonal output. These correlations persist across network sizes, consistent with a scale-invariant structural organization of cortical connectivity. We hypothesize that dendritic spines provide a structural substrate for network-level cooperation, by enabling distributed cortical activity to propagate through parallel synchronous chains, potentially amplified by nonlinear voltage responses at the spine and cellular level. These findings suggest that dendritic spines are key structural mechanisms for neuronal ensembles, attractor dynamics, and pattern completion.

Introduction

Dendritic spines, small protrusions emerging predominantly from the dendritic trees of cortical excitatory neurons, are widely hypothesized to broaden a neuron’s sampling capacity by increasing the number and diversity of its synaptic inputs (Yuste, 2011). This ‘connectivity and diversity hypothesis’ suggests that spines are the primary structural mechanism for expanding a neuron’s reach within the network. Uncovering the network logic of these structures, however, requires synaptic-resolution connectivity data across massive large scales. To investigate this, we utilized the MICrONS electron microscopy-based reconstructed volume, a $\sim 1 \text{ mm}^3$ reconstruction of mouse primary V1 cortex comprising a total of ~ 337 million synapses (MICrONS Consortium, 2025).

Historically, the morphological division between dendritic spines and shafts has been tightly coupled to distinct functional roles and presynaptic origins. It is well established that the overwhelming majority of dendritic spines are actively engaged in synaptic transmission, with only a marginal fraction (approximately 3.6%) lacking a distinct synaptic contact (Arellano et al., 2007). At the subcellular level, there is a pronounced targeting bias: the vast majority of excitatory inputs are directed to dendritic spines (Peters et al., 1991), whereas inhibitory inputs predominantly innervate dendritic shafts (Megías et al., 2001; Kwon et al., 2019). However, the dendritic shaft is not an exclusively inhibitory domain. Recent evidence highlights that also for excitatory synapses, 25%–30% of postsynaptic sites are located directly on the dendritic shaft, and these excitatory shaft synapses are proposed to constitute a unique, less stable synaptic population that significantly contributes to neuronal signalling and structural flexibility (Fanutza et al., 2025).

While these past studies have illuminated the localized properties of different synaptic compartments, translating these subcellular rules into a cohesive, network-wide functional logic requires anatomical analysis on a significantly larger scale. Spine anatomy has been well-characterized on the level of broad classes of neurons, such as excitatory

versus inhibitory populations. But it remains unclear how spines enable neurons to find their individual, highly specific roles within the circuit. Their marked differences in functional properties stem, at least partially, from a great diversity of connectivity, such as long-tailed degree distributions (Brunel, 2016; Piazza et al., 2025). Additionally, cortical connectivity has been characterized as highly structured with a tendency for assortativity, i.e., the tendency that neurons that are similar in some regard preferentially wire together (Ko et al., 2011; Cossell et al., 2015). One such example is the so-called rich-club organization (Van Den Heuvel and Sporns, 2011). Intuitively, we can speculate how spines play a role in all of this. They may increase the number and diversity of neurons a cell can contact, enabling the implementation of specialized wiring diagrams with high node-level diversity. But structural support for this idea has been elusive. As a starting point, we correlate the spine anatomy of neurons with their precise positions in the wiring diagram.

To ensure that our structural insights reflect biological ground truth rather than automated segmentation artifacts, we grounded our primary connectivity analyses in a proofread local micro-column (Schneider-Mizell et al., 2025). By relying on a verified subcircuit—where human annotators have corrected common automated errors such as fragmented “splits” or incorrectly joined “mergers”—we ensure that the identified synaptic partnerships accurately reflect biological ground truth rather than segmentation artifacts, providing a reliable foundation for analysis.

Leveraging the connectomic mapping of hundreds of manually proofread cells and millions of synapses, we hereby provide quantitative structural evidence supporting established connectivity differences between excitatory and inhibitory neurons, while proposing a novel functional role for dendritic spines. In Figure 1, we compare the incoming synaptic density across cell types, specifically contrasting spine and shaft distributions for both excitatory and inhibitory neurons. Figure 2 provides direct evidence corroborating the connectivity and diversity hypothesis (Yuste, 2011) by showing that higher dendritic spine density serves as a structural predictor of the number of synaptic inputs and the diversity of unique presynaptic partners across both excitatory and inhibitory populations. In Figure 3 we demonstrate that the preference of excitatory axons for spines increases with distance from the soma of the presynaptic neuron. This targeting rule has first been noticed by Loomba et al. (2022), based on a smaller dataset. Moreover, we also found an additional correlation where the more outgoing synapses an excitatory axon has, the higher the chances they target spines. This indicates that spines also increase the quantity and diversity of the outgoing connectivity potential. This is however not the case for inhibitory axons, thus highlighting an ultrastructural difference between excitatory and inhibitory axons. Figure 4 links this structural spine-preference to network-wide collaboration; supported by a graph-theoretical proof (see Appendix), we argue that spines—which preferentially receive input from highly broadcasting “hub” neurons with extensive axons—collectively serve as the structural backbone of network cooperativity. In Figure 5, we address potential dataset limitations through a conceptual sub-sampling experiment, showing that our conclusions remain robust even within progressively smaller cortical volumes.

Together, these findings extend the “connectivity and diversity” hypothesis for spines in a new direction. Dendritic spines increase the number of possible synaptic partners, but also play an active role in how they are selected. In this regard they appear to form a structural substrate for network-level collaboration: they preferentially receive input from broadly projecting excitatory neurons and participate in shared-input architectures. Decades of research have described various organizing principles of cortical connectivity—such as common neighbours (Perin et al., 2011), assortativity (Ko et al., 2011; Cossell et al., 2015), and rich-club architectures (Van Den Heuvel and Sporns, 2011)—and computational research has extensively demonstrated their profound functional impact. Here, we reveal that dendritic spines structurally underlie all of these phenomena. This convergence leads to the possibility that understanding the local rules governing spine growth and targeting could advance our understanding of the connectome.

Results

Differences in connectivity among excitatory and inhibitory neurons

Our analysis evaluates this dataset across two distinct topological scales. The first is a densely reconstructed local micro-column ([Schneider-Mizell et al., 2025](#)), containing $\sim 146,000$ intrinsic synapses exclusively connecting a highly curated subset of 1,298 fully reconstructed and manually identified pre- and post-synaptic neurons within the subcircuit (filtered from an initial set of 1,352; see [Methods](#)). This micro-column is situated at the center of the VISp region, spanning a $100 \times 100 \mu\text{m}$ lateral area across the entire cortical depth.

The second scale is a large-scale ‘global’ network that captures the extensive connectivity between the micro-column and the surrounding cortical tissue, comprising ~ 4.6 million incoming synapses projecting onto the micro-column from the broader $\sim 1\text{mm}^3$ volume (and outside of it), alongside ~ 1 million outgoing synapses originated from the micro-column and projecting to the rest of the 1mm^3 volume. For all these synaptic connections a prediction of its ultrastructural target—spine or shaft—is estimated to reach 95% accuracy ([Pedigo et al., 2026](#)).

The local micro-column represents a uniquely high-fidelity subset of this volume because it is a complete axon-proofread dataset. While automated segmentation algorithms can reconstruct large-scale volumes, they remain prone to “splits” (fragmented processes) and “mergers” (incorrectly joined processes). To mitigate these errors, the MICrONS team utilized human annotators to review and correct the axonal arbors within this subcircuit. This manual verification is particularly important for axons, which are significantly thinner and more morphologically complex than dendrites, making them highly susceptible to automated errors. By utilizing an axon-proofread dataset, we ensure that the identified synaptic partnerships accurately reflect biological ground truth rather than segmentation artifacts, providing a reliable foundation for our analysis. Furthermore, despite its localized scale, this verified subcircuit was shown to serve as a faithful representative of the broader VISp region in both its cell type composition and their relative proportions ([Schneider-Mizell et al., 2025](#); [Zhang et al., 2026](#)).

The distinction between these two topological scales is governed by specific analytical “rules” that ensure biological validity across varying levels of reconstruction detail. The local micro-column serves as our primary foundation for connectivity analysis; because it is axon-proofread, we can construct a high-confidence adjacency matrix where every synapse is traced back through its verified axonal arbor to its specific pre-synaptic origin. However, to address questions requiring greater statistical scale, we utilize the ‘global’ network under two conditions. First, when analyzing total synaptic input richness (e.g., [Fig. 1](#)), we leverage the ~ 4.6 million incoming synapses. This is justified by the relative morphological stability of dendrites—which are less prone to automated segmentation errors than axons—and the fact that this analysis requires only the accurate detection of a synapse on a known post-synaptic target, rather than the identity of the pre-synaptic partner. Second, when quantifying axonal output (e.g., [Fig. 3](#)), we utilize the ~ 1 million outgoing synapses. Here, we rely on the proofread axons within the micro-column to calculate total synaptic release, a metric that remains robust regardless of the specific identities of the target neurons residing in the surrounding 1mm^3 volume. By applying these constraints, we maintain the structural rigor of a proofread subcircuit while exploiting the extensive statistical power of the broader cortical network.

[Figure 1](#) illustrates the distinct synaptic connectivity of excitatory and inhibitory neurons in a densely reconstructed cortical circuit from mouse V1 ([Fig. 1A](#)). The analysis is based on a proof-read micro-column ([Fig. 1A](#), colored) derived from the $\sim 1\text{mm}^3$ MICrONS dataset of mouse V1 ([Fig. 1A](#), grey box). The micro-column initially contained 1,352 manually identified neurons, which were filtered to a final dataset of 1,298 neurons with fully resolved structures (1,139 excitatory and 159 inhibitory; see [Methods](#)). These neurons are connected through $\sim 146,000$ intrinsic synapses (where both the pre and the postsynaptic neurons are in the sub-circuit), receiving a total of ~ 4.6 million incoming synapses from the surrounding tissue (from the 1mm^3 cut volume and, notably, also from areas outside of it, such as the thalamus) and forming ~ 1 million outgoing synapses onto the whole volume ([Fig. 1A](#)). Representative reconstructions highlight the striking morphological differences between the excitatory and inhibitory classes ([Fig. 1B](#)). Excitatory pyramidal neurons exhibit dendrites densely decorated with dendritic spines, whereas inhibitory interneurons display comparatively smooth dendrites with far fewer spines ([Fig. 1B](#)). Zoomed views of representative dendritic branches and corresponding EM sections illustrate that excitatory dendrites receive mostly spine synapses, whereas inhibitory dendrites are contacted primarily by shaft synapses.

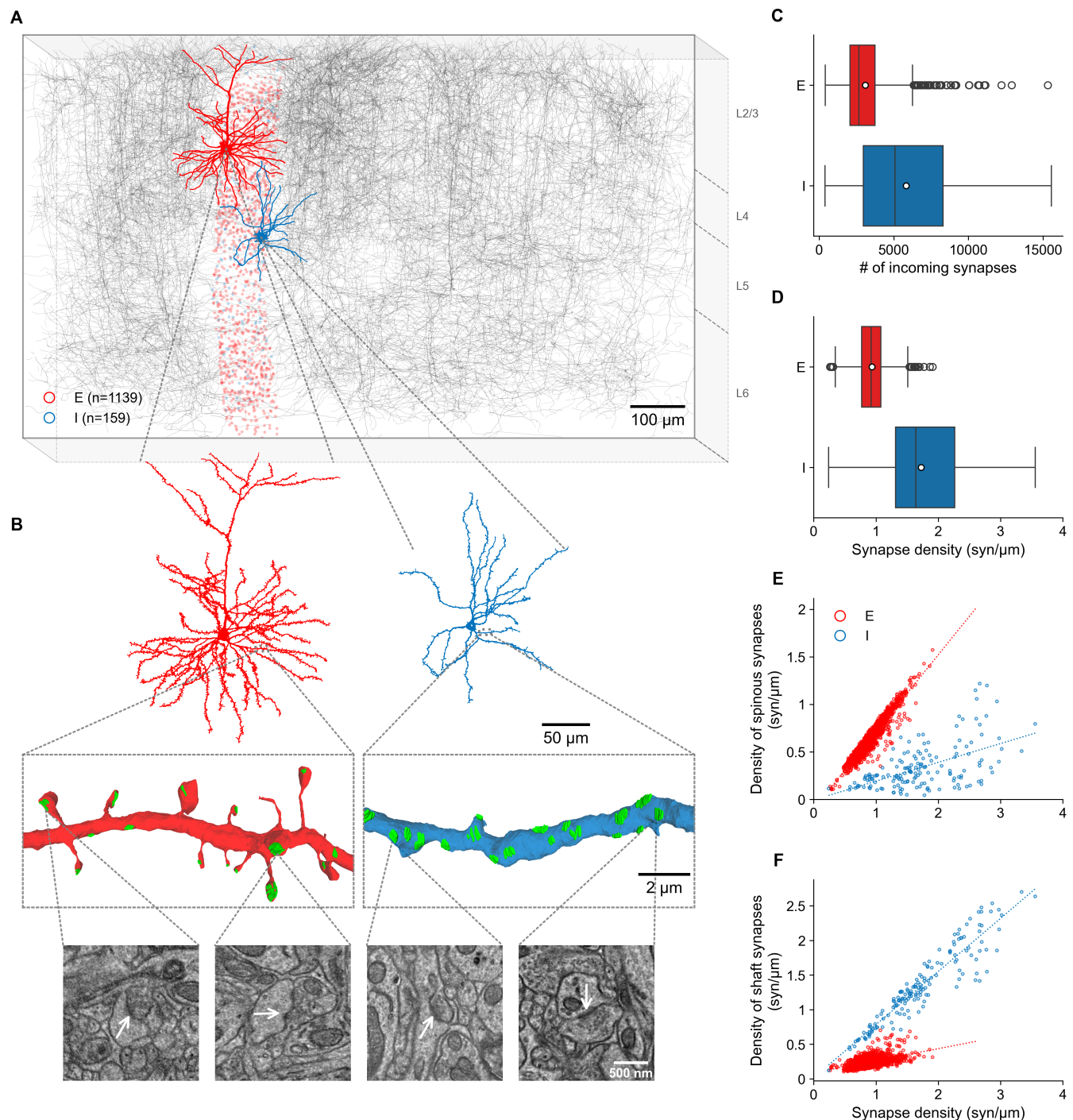


Figure 1. Distinct synaptic connectivity of excitatory and inhibitory neurons: excitatory neurons receive mostly spine synapses whereas inhibitory neurons receive more synapses on dendritic shafts. **A.** Gray network visualization of the $\sim 1 \text{ mm}^3$ mouse V1 circuit, derived from the MICrONS dataset dense reconstruction; 400 neurons were randomly sampled for this display. The proofread micro-column (colored) contains 1,298 neurons (1,139 excitatory, red; 159 inhibitory, blue) after basic structural filtering (see [Methods](#)) and connected by $\sim 146\text{K}$ intrinsic synapses, receiving a total of $\sim 4.6\text{M}$ incoming synapses from the whole volume, and making $\sim 1\text{M}$ outgoing synapses. **B. Top:** Zoom-in of two representative neurons from the micro-column shown in A: a layer 2/3 pyramidal neuron (red) and a L5 basket interneuron (blue); only dendrites are shown. **Middle:** Higher magnification views of representative dendritic branches from the neurons shown above. **Bottom:** Representative EM sections illustrating synapses contacting the dendrites shown above; arrows pointing at the post-synaptic density. **C–D.** Quantification of synaptic inputs for excitatory and inhibitory neurons within the whole volume; **C** shows the number of incoming synapses per neuron whereas **D** displays synapse density (synapses/ μm). In both panels, the box edges represent the 25th and 75th percentiles, the vertical line denotes the median, and the white circle indicates the mean. Whiskers extend to $1.5\times$ the interquartile range. Mean $\pm$ SD values: **C**, excitatory: $3,097 \pm 1,714$ vs. inhibitory: $5,828 \pm 3,507$; **D**, excitatory: 0.97 ± 0.25 vs. inhibitory: 1.81 ± 0.68 synapses/ μm . **E.** Density of spinous synapses as a function of total synapse density for excitatory and inhibitory neurons. Dotted lines represent linear regressions (excitatory: slope = 0.81 , $R^2 = 0.9$, $p < 0.001$; inhibitory: slope = 0.2 , $R^2 = 0.3$, $p < 0.001$). **F.** Density of shaft synapses as a function of total synapse density. Dotted lines represent linear regressions (excitatory: slope = 0.18 , $R^2 = 0.28$, $p < 0.001$; inhibitory: slope = 0.77 , $R^2 = 0.87$, $p < 0.001$).

Quantitative analysis of synapse number and density shows that inhibitory neurons receive more and denser synaptic inputs than excitatory neurons. On average, inhibitory cells receive almost twice as many incoming synapses, $5,828 \pm 3,507$ vs. $3,079 \pm 1,714$, respectively (Fig. 1C) and exhibit roughly two-fold higher synapse density along their dendrites, 1.81 ± 0.68 vs. 0.97 ± 0.25 synapses/ μm , respectively (Fig. 1D).

Previous ultrastructural and functional mapping studies have suggested distinct subcellular targeting preferences for excitatory and inhibitory neurons (Kwon et al., 2019), a bias that becomes strikingly evident at scale in the MICrONS dataset. In excitatory neurons, the majority (70%) of synaptic contacts occur on dendritic spines whereas the dendritic shaft is only sparsely decorated with synapses from the surrounding tissue; the variability of synapse density is almost completely explained by variability of spine synapses (Fig. 1E), indicating that connectivity expansion onto excitatory neurons occurs primarily through dendritic spines. This means that, in excitatory neurons with low spine density, the lack of spine synapses is not compensated by an increase in shaft synapses, even though there is more space on the shaft. Conversely, inhibitory neurons receive most (80%) of their synapses directly on the dendritic shaft; their synapse count scales almost entirely through shaft synapses (Fig. 1F), which increases steeply with synapse density. Together, these results strengthen and scale up earlier observations of compartment-specific spine versus shaft connectivity in excitatory and inhibitory cortical neurons (Chen et al., 2012).

These findings point to a fundamental principle of cortical circuits: excitatory neurons integrate inputs through spine-based compartments that may enable more selective and localized dendritic computations, whereas inhibitory neurons receive denser, likely less selective, shaft-based inputs that may perform dendritic computations more globally (see below and Discussion).

Dendritic spines density correlates with input number and diversity

Figure 2 shows that dendritic spine density is correlated with the richness and diversity of synaptic inputs to the respective neuron. Figure 2A illustrates two reconstructed pyramidal neurons showing the spatial distribution of their presynaptic partners originated within the local micro-column depicted in Figure 1. Presynaptic neurons are distributed throughout the surrounding network and include both excitatory (triangle) and inhibitory (circle) partners. Representative dendritic segments from excitatory and inhibitory neurons further illustrate the structural contrast between spiny and aspiny dendrites (Fig. 2B).

Across the population of pyramidal neurons, higher spine density is associated with increased local synaptic input originating from within the micro-column (Fig. 2C, top). The number of synapses increases by approximately a factor of 4.5, from 15.4 to 70.1, as spine density increases from 0.25 to 1.25 spines/ μm (Fig. 2C). For excitatory inputs this is expected from the overall increase in synapse count associated with increased spine density (Fig. 1E). Remarkably, we observed a similar increase in number of inhibitory inputs with increase in spine density, although inhibitory synapses largely target shafts ($\sim 80\%$, see Fig. S5), which means that their abundance could have been independent of spines (Fig. 2C, top, dotted line). For inhibitory presynaptic neurons, the number of synapses that they make per postsynaptic excitatory neuron increases by a factor of 4.6, from 32.6 to 150.5, as spine density increases from 0.25 to 1.25 spines/ μm (see Methods).

Interestingly, increase in spine density is also associated with a larger number of synapses made by individual inhibitory axons onto the postsynaptic cell (i.e., multiple connections). The average number of inhibitory synapses per axon per postsynaptic neuron increases from 2.1 to 3.6 synapses/axon with increase in spine density of the postsynaptic excitatory neuron (Fig. 2C, middle, dotted line). In contrast, excitatory axons typically form only a single synapse per axon over the target neuron, largely independent of spine density (Fig. 2C, middle, filled line).

The lower redundancy of excitatory compared to inhibitory inputs could be the statistical result of the lower count of local excitatory synapses, combined with the smaller population of potential presynaptic inhibitory partners. However, this means that the increase in excitatory synapse density resulting from higher spine density is directly translated into an increase in the number of distinct presynaptic partners, which rises from 13.7 partners to 58.8 excitatory partners as spine density increases from 0.25 to 1.25 spines/ μm (Fig. 2C, bottom). Thus, in excitatory neurons, dendrites with higher spine density receive synaptic input from a larger and more diverse set of presynaptic neurons.

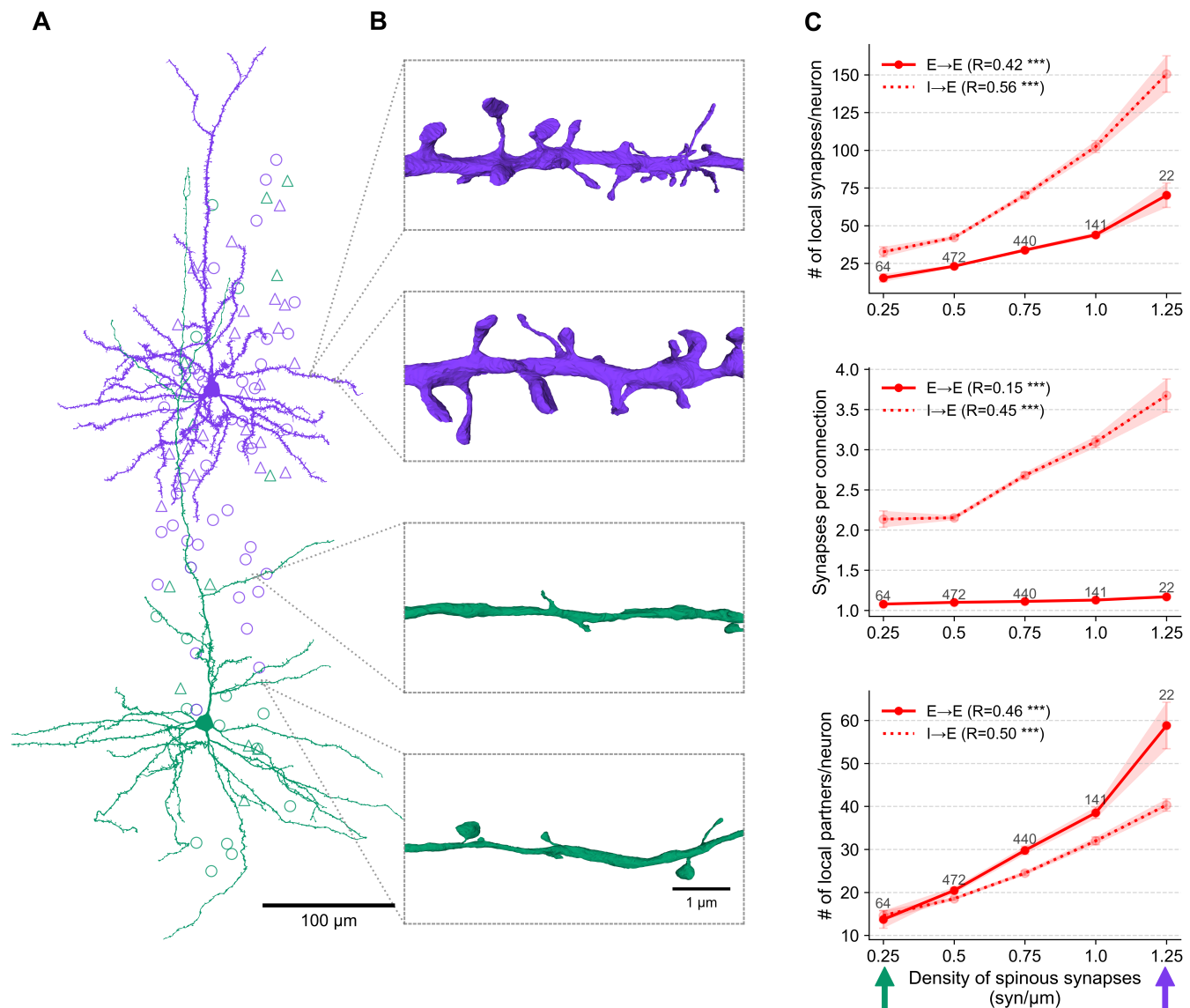


Figure 2. Spine density correlates with local synaptic input richness and diversity. **A.** Example reconstruction of layer 4 (purple) and layer 6 (green) cortical pyramidal neurons with two different spine densities, depicting the spatial distribution of their presynaptic partners' soma location within the micro-column shown in Figure 1A; triangles for excitatory and circles for inhibitory; only dendrites are shown. **B.** Representative dendritic segments from neurons shown in A, illustrating the contrast between densely and sparsely spiny dendrites of excitatory neurons of different types. **C. Top:** Relationship between number of local synaptic inputs per neuron, originating within the micro-column, and total spine density for excitatory neurons. **Middle:** As in top but for the number of synapses per connection (i.e., per presynaptic axon). **Bottom:** As in top but for number of distinct presynaptic partners per neuron. Solid lines represent excitatory-to-excitatory connections and dashed lines inhibitory-to-excitatory connections. Arrows at the bottom illustrate the spine density of the neurons in panel A.

Equivalent analyses across the population of inhibitory neurons reveal a different organizational principle (Fig. S1C; representative morphologies depicted in Fig. S1A–B). Although inhibitory neurons receive substantially more local synaptic input overall (the majority of neurons receive between 100–200 excitatory synapses across the observed spine-density range, with only 17 neurons receiving between 200–400 synapses). For inhibitory neurons, the number of inhibitory synapses/neuron remains roughly constant (56.9 synapses/neuron) and largely independent of spine density (Fig. S1C top, dotted line). Moreover, the scaling of multiple synapses from individual presynaptic axons diverges sharply between cell types. While excitatory neurons show a pronounced increase in multi-synaptic connections from inhibitory inputs, inhibitory neurons exhibit the inverse pattern: increased spine density is instead associated with a slight increase in multiple synapses formed by individual excitatory axons (E→I, peaking around

1.5 to 2 synapses per connection, Fig. S1C middle), whereas connections from individual inhibitory axons (I→I) show little to no increase. Furthermore, higher spine density in these neurons is specifically associated with a larger number of excitatory presynaptic partners (increasing from an average of ~85 to ~139), while the number of inhibitory presynaptic partners remains approximately constant (21 partners per neuron, Fig. S1C, bottom). This relationship holds true even when normalizing for total dendrite length (Fig. S2), indicating that these overarching trends are not merely a byproduct of larger dendritic trees but represent a specific connectivity logic.

These findings refine our understanding of how dendritic structures mediate connectivity. In excitatory neurons, spines clearly serve to expand network sampling. In inhibitory neurons, the architecture is different: shaft synapses are the main driver of input diversity (Fig. 1); yet spines still maintain a measurable impact on connectivity by facilitating more (excitatory) partners.

Comparison of presynaptic partner counts for excitatory and inhibitory neurons (Fig. 2C and Fig. S1C, bottom) together with the results of Figure 1 reveals a connectivity principle in local cortical circuits. In both cell classes, dendritic spine density acts as a structural predictor for the richness and diversity of synaptic input, providing quantitative validation for the connectivity and diversity hypothesis (Yuste, 2011). In excitatory neurons, this relationship is particularly potent, as spine density strongly predicts both the number and diversity of local presynaptic partners. While inhibitory neurons receive denser shaft-based inputs and more presynaptic partners overall, they still follow this “spine-for-diversity” logic specifically for their excitatory inputs. Although this dependence on spine density corresponds to a roughly 4-fold increase in connections for excitatory neurons, the consistent trend across both populations suggests that spines are the primary morphological tool used by cortical neurons to broaden their sampling of the surrounding network.

Length of excitatory axons correlates with synaptic output number and spine targeting

After describing and characterizing a large variability of spine vs. shaft preference with respect to origin of the input synapses, we next considered neuronal outputs. Specifically, we examined the relationship between axonal extent and synaptic output in reconstructed cortical axons. Previous topological analyses of neocortical microcircuitry have demonstrated that neurons with extensive, highly branched axons effectively function as dense “outgoing hubs” that broadly distribute signals across the network (Gal et al., 2017). Our volumetric data analysis provides direct, large-scale structural evidence for this architecture. Representative examples of reconstructed axons are shown in Figure 3A–C; a long-range L5-IT pyramidal neuron forming large numbers of synapses (Fig. 3A); L6-CT pyramidal neuron with a shorter axon and with much less output synapses (Fig. 3B); and an axon of a basket cell forming dense output synapses (Fig. 3C).

Analysis of the connectivity of excitatory axons, originating within the micro-column but projecting both in and outside of the micro-column, shows that the total number of outgoing synapses correlates linearly with axon length (Fig. 3D, left; Pearson $R = 0.88$), with a slope of 7 synapses/100 μm (one synapse per 14 μm) for excitatory axons, in agreement with Loomba et al. (2022) and Piazza et al. (2025). Interestingly, and in agreement with Schmidt et al. (2017) and Loomba et al. (2022), the ultrastructural target of these output synapses changes systematically with increasing synaptic output number; the fraction of synapses onto dendritic spines increases with the number of outgoing synapses (Fig. 3D, right; Spearman $\rho = 0.69$). Namely, excitatory neurons with longer axons not only make more output synapses in total, but also exhibit a stronger preference for spine targeting. Using the micro-column where the identification of both pre- and post-synaptic neurons is known, we next examined whether the relationship between synaptic output and spine targeting depends on postsynaptic cell (E or I) type. Excitatory-to-excitatory (E→E) connections show a strong increase in spine targeting with the number of outgoing synapses (Fig. 3H, left; $\rho = 0.55$). In contrast, excitatory-to-inhibitory (E→I) connections show a weaker dependence (Fig. 3H, right; $\rho = 0.22$).

In this context, it should be noted that we use the terms “spine targeting” or “axonal preference for spines” as a descriptive shorthand for our observations; we do not imply that the mechanisms underlying the observations lie exclusively on the presynaptic side, nor do we discount the active roles of the spines in the formation of a synapse. To determine whether distance-dependent spine preference reflects spatial organization along the axon, we quantified spine targeting as a function of axonal path distance from the soma. Across the excitatory axons, the

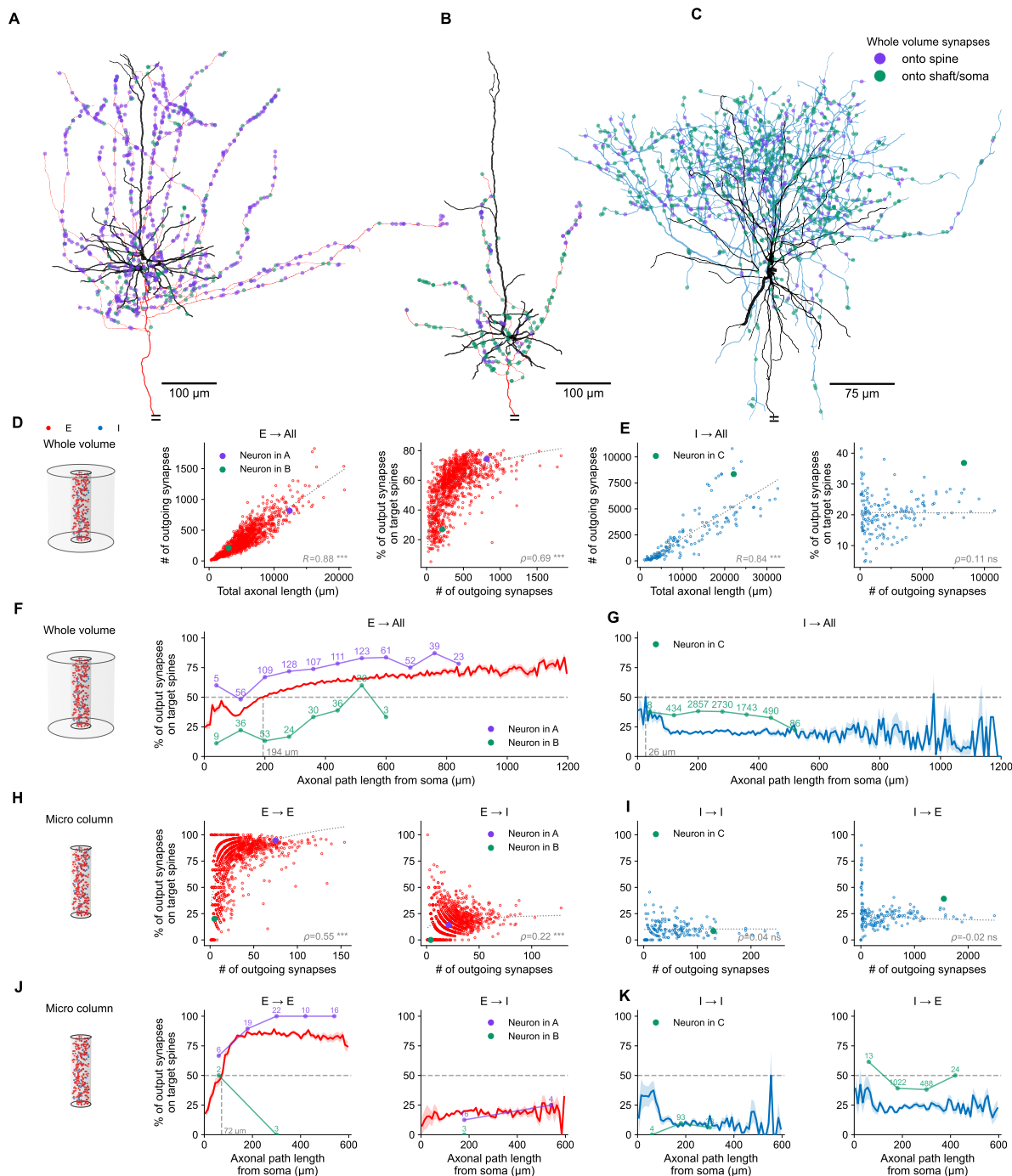


Figure 3. Spine targeting increases with axonal distance from soma in excitatory but not inhibitory neurons. **A.** Reconstructed L5-IT pyramidal neuron with an extensive axonal arbor and a large number of outgoing synapses. Dendrites in black and axons in red, synapses onto spines in purple and shaft/soma synapses in green. **B.** Reconstructed L6-CT pyramidal neuron with a shorter axon and fewer outgoing synapses. **C.** Reconstructed L4 basket interneuron. Dendrites in black and axons in blue. For visualization clarity, 10% of outgoing synapses are randomly sampled and colored as in A–B. **D.** **Left:** Number of excitatory axonal synapses versus total axonal length; dotted line indicates an average density of 7 synapses per 100 μ m of axon ($R = 0.88$, $p < 0.001$). Neurons in A and B are highlighted in purple and green, respectively. **Right:** % of excitatory synapses onto target spines as a function of the total number of outgoing synapses ($p = 0.69$, $p < 0.001$). **E.** As in D for inhibitory neurons. **Left:** Number of synapses versus axonal length ($R = 0.84$, $p < 0.001$), average density of 25 synapses per 100 μ m. The example neuron in C is highlighted (green). **Right:** % of synapses onto target spines showing no significant dependence on the number of outgoing synapses ($p = 0.11$, $p = 0.15$). **F.** Spine targeting as a function of path distance along excitatory axons. Population average (red) and individual neurons from A (purple) and B (green) are shown, with the number of synapses per bin indicated above respective curves. The horizontal dashed line marks the distance at which 50% of synapses are made onto spines. **G.** Same as F for inhibitory axons. Spine targeting remains low and largely distance-independent. **H.** **Left:** Excitatory-to-excitatory (E→E) connectivity shows a strong increase in spine targeting with outgoing synapses ($p = 0.55$, $p < 0.001$). **Right:** Excitatory-to-inhibitory (E→I) shows a weaker relationship ($p = 0.22$, $p < 0.001$). **I.** Same as H for inhibitory axons. I→I and I→E show no relationship ($p \approx 0$, ns). **J.** Same as F for the micro-column. **Left:** E→E shows similar increase. **Right:** E→I shows a lower baseline that remains shallow across axonal path lengths. **K.** Same as G for the micro-column. **Left:** I→I and **Right:** I→E both show spine targeting remains low and independent of axonal path distance. Statistical significance for linear correlations was determined using Pearson's R or Spearman's rank correlation (ρ) as indicated.

fraction of synapses onto spines transitions from a low baseline near the soma to a spine-dominated regime at larger distances (Fig. 3F, red curve). This progression exhibits a transient dip around 100 μm ; analysis of the underlying synapse distributions reveals that this dip is driven by a dense, localized zone of shaft-directed synapses onto nearby inhibitory interneurons (Fig. S4B).

Notably, spine targeting transitions from a mixed (shaft/spine) regime near the soma to a spine-dominated regime at larger distances (Schmidt et al., 2017; Loomba et al., 2022). We found that, at the population level, the distance at which spine targeting reaches 50% is $\sim 200 \mu\text{m}$ (Fig. 3F, dashed line, see Methods), which provides a characteristic length scale for this transition. However, individual neurons exhibited substantial diversity. A large subset of neurons preferred spines all along their reconstructed axon, remaining continuously above this 50% threshold (37%), while a very small fraction never reached this threshold (0.1%). For the remaining neurons that did have this transition distance, the crossing occurred at a median distance of 83 μm (IQR: 49.1–114.5 μm), with distinct distributions across different morphological cell types (Fig. S7, see Methods).

Given that local inhibitory targets primarily receive synapses on their dendritic shafts while distal excitatory targets integrate inputs via spines, this spatial transition is naturally related to the distance-dependent shift in target cell-type preference described in prior studies (Schmidt et al., 2017; Bodor et al., 2025). However, we demonstrate that a shift in postsynaptic cell type alone cannot fully account for this phenomenon. Crucially, when the local inhibitory population is removed and the analysis is strictly confined to the excitatory-to-excitatory (E→E) subnetwork, the localized dip disappears, and the progressive increase in spine targeting remains highly pronounced (Fig. 3J, left). In contrast, for E→I connections, as well as all inhibitory outputs (I→I and I→E), spine targeting stays at a lower baseline and remains largely independent of axonal path distance from the soma (Fig. 3J, right; Fig. 3K). This indicates that the progressive preference for dendritic spines is a subcellular organizing principle of excitatory axons, rather than a mere by-product of cell-type targeting. In fact, the result might indicate the reverse: that the shift in target cell-type preference is a side effect of (or implemented as) a shift in spine preference. In that case, cells with an overall lower preference for spines would also exhibit a lower preference for excitatory targets. We found that this is indeed the case. When examining the local connectivity within the micro-column, an excitatory neuron’s preference for targeting other excitatory partners (quantified as the ratio of E to E+I partners) is only weakly correlated with its total number of outgoing synapses (Spearman $\rho = 0.20$). However, this preference is robustly predicted by the neuron’s overall preference for targeting dendritic spines (Spearman $\rho = 0.56$; Fig. S6).

The observed increase in the probability, $p(x)$, of spine targeting with axonal path length from soma can be captured by a simple model:

$$p(x) = \rho_{\infty} \cdot \frac{x}{x + d_0} \quad (1)$$

where x is the axon’s path distance from soma to the synapse (measured in μm), ρ_{∞} is the asymptotic spine-targeting probability, and d_0 is a characteristic length scale governing the transition from the initial regime which is biased towards the dendritic shaft to the asymptotic regime, dominated by spines.

We tested the fit of this equation to our data; the results are shown in Figure S4. This showed two basic results: first, the fit is robust against the number of spatial bins used; second, the model works best at the population level. The natural variation and random branching within a single axon introduce greater noise and make the fit much less reliable compared to an entire population. We found that the asymptotic spine-targeting probability (ρ_{∞}) remains consistently high (ranging from 0.82 to 0.93), but the characteristic length scale (d_0) varies drastically depending on the target population. For these presynaptic excitatory neurons, the characteristic scale, d_0 is 124 μm when considering all available targets across the entire volume (E→All) and 148 μm for all target neurons within the micro-column (E→E&I). This scale decreases to 28 μm when considering strictly excitatory targets within the micro-column (E→E) (Fig. S4). This indicates that the transition to a spine-targeting preference happens over a much shorter distance when axons are restricted to an excitatory-only network.

On the level of individual neurons we found substantial variability in the overall fraction of spine synapses targeted by their axon (Fig. 3F, purple vs. green), but the increase of the probability to target a spine with path distances from soma still holds (Eq. 1). This indicates that the population-level increase is not the consequence of averaging over axons of different lengths. Neither is the increase in spine preference for longer axons explained by

longer axons simply having more mass at longer distances (Fig. S8). To determine if distance alone could account for this trend, we applied our distance-dependent simple model (fitted to the population average, Fig. S4) to the empirical synapse distances of each individual neuron. By calculating a predicted global spine preference based purely on the spatial distribution of each neuron's synapses, we found that distance-based rules alone cannot replicate the wide cell-to-cell variance or the full magnitude of the increase in spine preference seen in neurons with many outgoing synapses. In other words: higher spine preference for longer axons and higher spine preference for distal axon segments are two separate, concurrent phenomena (see [Discussion](#)).

Length of inhibitory axons correlates with synaptic output number but not spine targeting

In contrast to excitatory axons, inhibitory axons exhibit a different organization principle. Although axonal length strongly correlates with the number of outgoing synapses, with an average of 25 synapses/100 μm (one synapse/4 μm , Fig. 3E, left; Pearson $R = 0.84$), the fraction of synapses onto spines does not depend on the number of axonal synapses (Fig. 3E, right; Spearman $\rho = 0.11$). Consistent with this, spine targeting of inhibitory axons remains low and largely independent of path distance along the axon (Fig. 3G), indicating the absence of a distance-dependent transition from shaft-to-spine preference. Thus, inhibitory axons do not exhibit the progressive shift toward spine targeting observed in excitatory axons.

When analyzing the micro-column subset, in inhibitory neurons, no significant relationship is observed for either inhibitory-to-inhibitory (I→I) or inhibitory-to-excitatory (I→E) connections (Fig. 3I), reinforcing the distinct targeting rules governing inhibitory connectivity. Further unpacking these divergent strategies, we find that the capacity to target spines directly drives the expansion of outgoing connectivity for excitatory neurons (Fig. S5). Their axons appear to establish a baseline supply of shaft-targeting synapses, with variations in overall output density determined almost completely by the addition of spine-targeting synapses. Inhibitory neurons operate under a fundamentally different logic; while their overall density of outgoing synapses is highly variable, they maintain a relatively constant spine-to-shaft ratio of approximately 20% (Fig. S5), scaling their output almost exclusively through shaft-targeting synapses.

To ensure that the observed relationship between axonal extent and spine targeting is not an artifact of incomplete axonal reconstruction, we performed a control analysis restricted to a subset of the most rigorously proofread axons, both inside and outside the micro-column (Fig. S3). A potential confound in large-scale connectomic datasets is that partially reconstructed axons might artificially appear both short and uniquely biased in their synaptic targeting, thereby driving a spurious population-level trend. To rule out this possibility, we limited our analysis strictly to a highly curated subset of axons with the highest level of manual proofreading (referred by the Allen Institute as “axon_fully_extended”, see [Methods](#)). This subset consists of 272 excitatory and 7 inhibitory neurons from within the micro-column, alongside an additional 42 excitatory and 126 inhibitory fully proofread neurons located outside the micro-column. These fundamental trends remain robust within this combined group. Total synaptic output continues to correlate linearly with total axonal length, and the fraction of spine-targeted synapses for excitatory axons still systematically increases with both total synaptic output (Fig. S3A) and path distance from the soma (Fig. S3C). This suggests that the progressive transition toward dendritic spines reflects a genuine biological feature of extended excitatory axons, further supported by the results in different cortical systems reported in [Loomba et al. \(2022\)](#) and [Schmidt et al. \(2017\)](#).

Together, these results reveal principles of cortical excitatory connectivity: excitatory axons with a greater total number of outgoing synapses exhibit a higher overall tendency to target dendritic spines, and this spine targeting also increases in a distance-dependent, saturating manner as the axon extends from the soma. Because distance alone cannot fully account for the higher spine targeting observed in neurons with extensive synaptic outputs, these likely represent concurrent but distinct structural phenomena. The spatial transition from early shaft-biased targeting to a spine-dominated regime is well captured by a simple model (Eq. 1) defined by a single transition scale d_0 . In contrast, inhibitory neurons do not exhibit such a transition, indicating fundamentally different structural rules for excitatory and inhibitory axonal targeting.

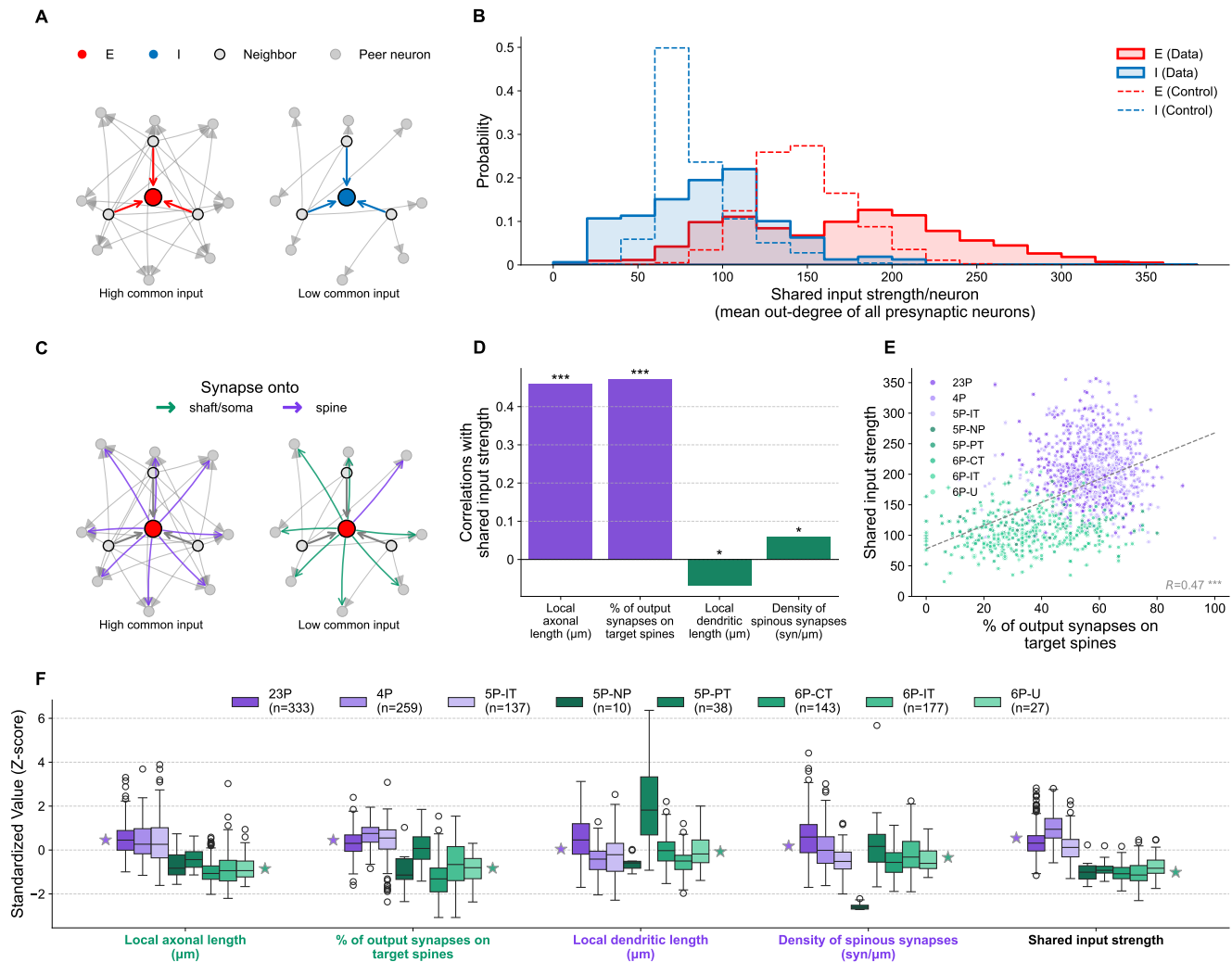


Figure 4. Shared-input innervation correlates with spine targeting. **A**, Schematic illustration of the sharing property: excitatory (E, red) and inhibitory (I, blue) neurons each receive input from 3 common presynaptic neighbours (gray circles with black outline) evaluated against surrounding peer neurons (solid gray circles). In the E-neuron case, its 3 presynaptic partners are high-out-degree nodes projecting broadly across the network, whereas the presynaptic partners of the I-neuron do not. Consequently, the E-neuron inherits a high tendency to share input with the rest of the network (high sharing neuron). **B**, Distribution of the shared input strength/neuron in the whole population. The excitatory (solid red) and inhibitory (solid blue) populations are compared against null distributions from 100 degree-preserving “blocked” configuration (CFG) models (dashed lines). Excitatory and inhibitory networks exhibit significantly distinct sharing properties (Mann-Whitney $U = 152,850.5$, $p < 0.001$; Kolmogorov-Smirnov $D = 0.55$, $p < 0.001$). **C**, Schematic illustrating the relationship between shared input strength and spine targeting. *Left*: a highly sharing neuron preferentially targets dendritic spines (purple arrows). *Right*: low sharing neuron tends to target the dendritic shaft (green arrows). **D**, Correlation between the neuron’s shared input strength and four features: the two left panels in purple display output features, while the two right panels in green display input features. **E**, Correlation between the neuron’s shared input strength and its spine preference (Pearson $R = 0.47$, $p < 0.001$). **F**, Z-scores of the features shown in E across different excitatory cell types. 2/3P, L4P, and 5P-IT populations (purple) exhibit the highest shared input strength and spine targeting preferences; deep-layer neurons and cortical output neurons (in green) display lower shared input strength and less spine-targeting preference. The purple and green stars indicate the mean per feature for all cell types in their respective colors.

Shared-input innervation is correlated with spine targeting

The organization of cortical circuits relies heavily on higher-order connectivity motifs to shape network function. A key principle in this domain is the ‘common neighbour’ rule (Perin et al., 2011), which established that the probability of a synaptic connection between cortical neurons is strongly predicted by the number of connections they share. While this collaborative architecture was originally identified functionally in cortical slices (Song et al., 2005; Perin et al., 2011), its underlying ultrastructural logic across massive anatomical scales has remained unexplored. We therefore next investigated the structural rules governing shared presynaptic input across the whole dataset.

Towards this end, we define “shared input strength”—a metric designed to measure the collaborative nature of each postsynaptic neuron, based exclusively on its incoming connectivity. This metric can be understood through

two distinct interpretations, illustrated using the schematics in Figure 4A. The first is a direct calculation of the number of common neighbours. For any given neuron we ask: how many presynaptic partners does it share with the rest of the network? To calculate this systematically, we compare the target neuron against every other neuron in the population. For example, if we take the red neuron and a peer neuron that is not directly connected to it (a gray circle, Fig. 4A), we test for shared inputs. Thus, whether a pair of neurons are directly connected to one another is irrelevant to this calculation. While the pair might happen to share a synapse, the shared input strength metric focuses strictly on the intersection of their presynaptic inputs—asking how many common neighbors they have in total, rather than looking at their own mutual relationship. As shown, one of the three “common neighbors” (gray circles with black outline providing direct synapses to the red and blue neurons) connect to both, thereby contributing to the score. By summing these shared partners across all possible pairs in the network and normalizing the result by the neuron’s in-degree (number of incoming connections it has), we ensure a metric that isn’t biased toward neurons that simply receive a high volume of synapses.

Essentially, this shared input strength value tells us how much a neuron “collaborates” or “shares” its input space with its peers. All of this is determined without considering the neuron’s outgoing connections at all; rather, this metric is exclusive to incoming connections only. As Figure 4A demonstrates, the red neuron exhibits a high degree of common input compared to the blue one. The red neuron is embedded in a more interconnected web; if we were to perform the systematic calculation described above, we would find that it shares significantly more inputs (with the rest of its network) than the blue neuron does.

This leads to a second interpretation of the “shared input strength” metric that is functionally and mathematically identical (see [Appendix](#)). Instead of counting every shared neighbor across the network, we can simply examine the presynaptic partners themselves. If the neurons providing input to our target node are “broadcasters”—namely high-out-degree nodes that project to a vast portion of the network—then the target neuron will inherently share more input with the rest of the network.

These two perspectives represent the same structural principle: a high propensity for sharing, measured by the average number of common neighbours, is mathematically equivalent to receiving inputs from high out-degree nodes. Consequently, a neuron with high common-input receives input from highly divergent, “broadcasting” partners. To align these concepts (particularly regarding the mathematical proof in the [Appendix](#)) it is important to clarify that while ‘out-degree’ is an abstract graph-theory term, it is tightly coupled to the outgoing synapses in our analysis. Because connecting to many unique target neurons (high out-degree) inherently requires forming a large number of outgoing synapses, out-degree serves as a direct structural proxy for synaptic output. Ultimately, a neuron’s shared input strength is driven by presynaptic partners that emit a vast number of synapses across the circuit.

Figure 4 shows the shared input strength analysis in the micro-column, using the second interpretation discussed above (Fig. 4B). We first observe that input sharing is overall more substantial among E-neurons (Fig. 4B, red) than I-neurons (Fig. 4B, blue; Mann-Whitney $U = 152,851$, $p < 0.001$; Kolmogorov-Smirnov $D = 0.55$, $p < 0.001$). However, the most prominent characteristic of the E-population is the structure of its variability compared to a degree-preserving configuration (CFG) control model. E-neurons exhibit a significantly wider distribution of shared input strengths than expected by chance (Kolmogorov-Smirnov $D = 0.33$, $p < 0.001$; Mann-Whitney $U = 79,764,135$, $p < 0.001$). To ensure a rigorous baseline, This CFG model shuffles each block of the matrix (i.e., EE, EI, IE, II) separately. By preserving the degree distributions within specific sub-circuits rather than shuffling the entire network as a single piece, we establish a more constrained and robust null expectation. This comparison reveals that, while I-neurons share less than predicted by the respective null model (Kolmogorov-Smirnov $D = 0.24$, $p < 0.001$; Mann-Whitney $U = 1,436,913$, $p = 0.003$), the E-population is characterized by a highly heterogeneous sharing architecture.

To further contextualize the role of this collaborative structure within the excitatory population, we evaluated how shared input strength relates to other fundamental morphological and synaptic features (Fig. 4C). “High-common-input” excitatory neurons preferentially direct their outgoing projections (purple) toward dendritic spines, whereas low-sharing neurons primarily target shafts (green). Indeed, we found that a neuron’s tendency to share presynaptic inputs correlates most strongly with its preference for targeting dendritic spines and with its local axonal length (the total axonal length confined only to the micro-column), surpassing the correlations with the density of the neuron’s

spinous synapses, or its local dendritic length (Fig. 4D).

What is unexpected about this robust structural link between the sharing strength and the axonal length and the tendency to target spines is that the former is fundamentally a property of the input (dictated by the connectivity and out-degree of presynaptic partners), whereas the latter are strictly output features.

As illustrated in Figure 4D, simply being a neuron densely covered with spines (thus possessing a high density of spinous synapses) is insufficient, as such, for it to be a highly sharing neuron. Rather, the defining characteristic to be a sharing neuron is the neuron's tendency for targeting spines on its postsynaptic partners. This solidifies the specific structural link between collaborative input integration and spine-directed synaptic output. Furthermore, this indicates that neurons receiving inputs from neurons with long axons—and hence a preference for spines—have long axons and spine targeting preference themselves. Such a tendency for similar neurons to connect to each other is captured by the term *assortativity*. In this case similarity is based on out-degree, resulting in a form of “rich club” (Van Den Heuvel and Sporns, 2011), but also on axonal spine targeting. Specifically, the assortativity score based on the density of spine-targeting synapses on the axon was 0.28, and the one based on out-degree was 0.2 (not shown).

Figure 4E shows the relationship between the shared input strength and spine targeting. This scatter plot reveals a robust correlation ($R = 0.47$) and highlights an apparent structural bifurcation within the population. Indeed, the distribution of shared input strength in Figure 4E and in the histogram of Figure 4B both display two distinct subpopulations. This is confirmed statistically by applying Hartigan's Dip Test (Hartigan and Hartigan, 1985), which indicated strong evidence against unimodality (Dip statistic = 0.017, $p = 0.025$; see Methods). This result confirms that the data are likely multimodal. Notably, this structural divide adheres closely to established cell-type identities from the MICrONS dataset: we colored the high-sharing, spine-preferring population in purple (2/3P, 4P, and 5P-IT) and the remaining populations in green (5P-NP, 5P-ET, 6P-CT, 6P-IT, 6P-U). For clarity, we excluded the ‘unsure E’ ($n = 4$) and ‘WM-P’ ($n = 11$) E neurons from Fig. 4E–F.

To further contextualize the axonal strategies driving the spine preference shown in Figure 4E, we directly compared the densities of outgoing spinous versus shaft synapses across these specific subpopulations (Fig. S12). This whole-volume assessment specifically captures the outgoing synapses among the $\sim 10^6$ synapses that project outside the local micro-column. Building on the broad output scaling differences between excitatory and inhibitory neurons established earlier (Fig. S5), we found that the bifurcation in excitatory spine preference is driven by distinct subcellular targeting strategies. The highly collaborative populations (purple neurons in Fig. 4E) actively scale their output by predominantly targeting spines, clustering well above the spine-to-shaft identity line. Conversely, the deeper-layers low-sharing populations (Fig. 4E, green) exhibit an inverted strategy, preferentially directing their output toward dendritic shafts. This subtype-specific structural diversity is uniquely excitatory; all identified inhibitory subtypes uniformly restrict their targeting to dendritic shafts, regardless of their specific classification (Fig. S12, right).

To fully characterize these relationships, we expanded our analysis to evaluate the correlations between shared input strength and structural features across the full network (Fig. 4F, Fig. S10). Furthermore, we also performed this analysis solely on the excitatory sub-network (E-E connectivity; Fig. S9). Consistent with our main findings, output features of the neuron consistently exhibited the strongest correlations with sharing strength in both network contexts. Noteworthy is that in the E-E sub-network a neuron's spine density—an input feature—emerges as a substantially better predictor of its sharing strength than it is in the full network. Still, even in this E-E subnetwork, spine density remains secondary to the predictive power of spine-targeting output. Finally, to verify that the core trends observed are not driven primarily by the highly curated subset of completely reconstructed axons, analyses were repeated after excluding the neurons with the highest level of manual proofreading (consisting of 272 excitatory and 7 inhibitory neurons from within the micro-column, Fig. S11). Our results remained highly consistent within this partially axon-extended population, with output features continuing to serve as the strongest predictors of a neuron's shared input strength.

By comparing standardized values (Z-scores) for these metrics across all cell types, a clear layer-specific hierarchy emerged (Fig. 4F). In agreement with previous results (Pedigo et al., 2026), our visualization confirms that the purple-labeled populations exhibit the strongest preference for targeting dendritic spines alongside greater local

axonal lengths. Our novel connectivity analysis reveals that these same spine-preferring populations also exhibit the highest propensity for sharing presynaptic partners. In contrast, deeper layer neurons demonstrate both the expected reduction in spine targeting ([Pedigo et al., 2026](#)) and noticeably lower sharing properties.

This anatomical distribution aligns intuitively with the canonical flow of cortical information processing ([Douglas and Martin, 2004](#)): highly collaborative, spine-preferring connectivity is densely enriched in the primary input (Layer 4) and local processing (Layer 2/3 and L5-IT) layers, whereas layers traditionally associated with broader cortical output (Layer 6) participate less heavily in this specific, local sharing motif.

In summary, an unexpected outcome of this topological analysis is the apparent link between a neuron's input architecture and its output targeting. The observation that a structural *input* feature—a neuron's tendency to share its input with the network—correlates so closely with its *output* behavior (its preference for innervating dendritic spines) suggests a highly organized “input-to-spine/output-to-spine” organization rule (see [Discussion](#)).

Connectivity patterns are maintained at different circuit spatial scales

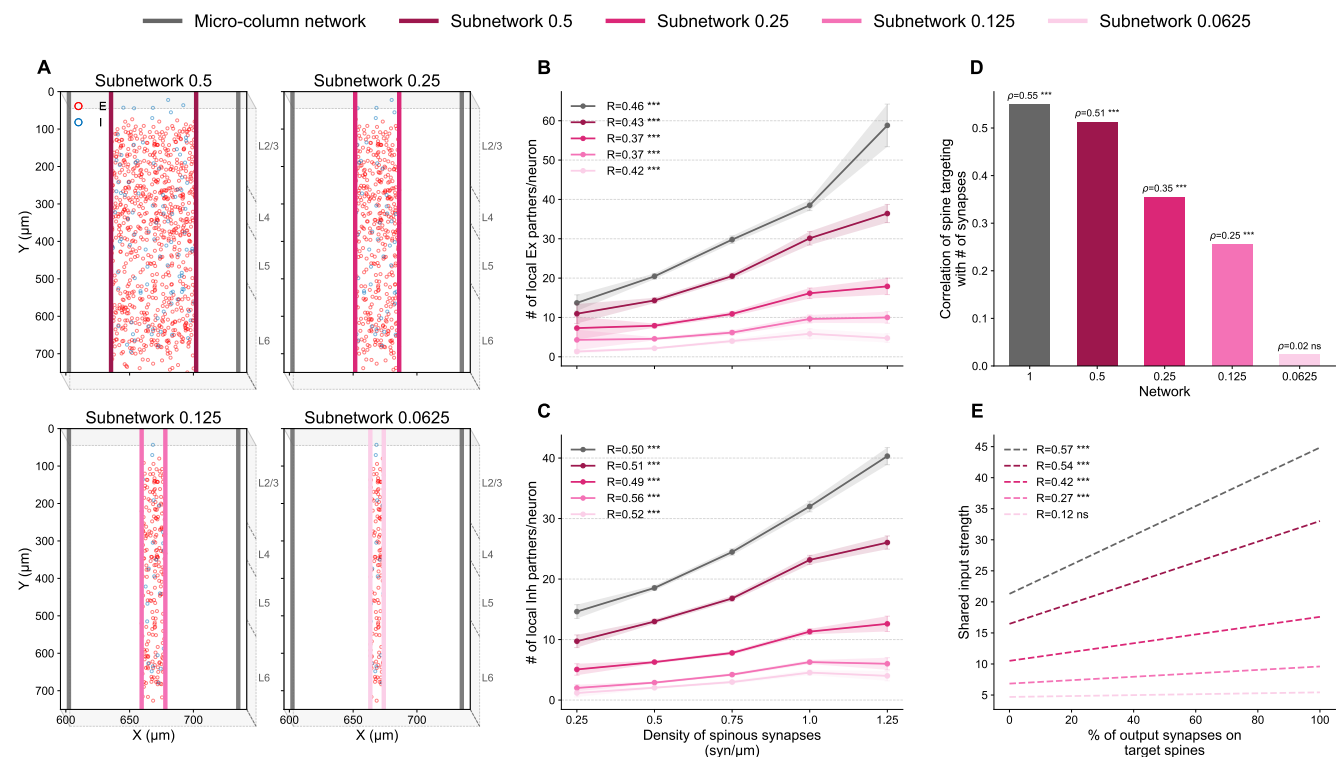


Figure 5. Scale-invariance of connectivity patterns across nested micro-columns. **A.** Illustration of a series of nested subnetworks from the original micro-column, whereby the X dimension is scaled by 0.5, 0.25, 0.125, and 0.0625, with Y (depth) and Z remaining constant. Connectivity and other features within each subnetwork are calculated based solely on neurons and synapses within the respective new boundaries. **B–C.** The number of unique local excitatory (B) and inhibitory (C) partner neurons for postsynaptic excitatory neurons plotted against spine density; same analysis as in Figure 2C. Color-coded lines correspond to the different subnetwork scales; grey for the original column. **D.** Spearman correlation (ρ) between the percentage of output synapses on target spines and the number of outgoing synapses for each subnetwork size, corresponding to the analysis in Figure 3D. **E.** Regression lines demonstrating the correlation between a neuron's shared input strength and its percentage of output synapses on target spines across subnetwork scales, corresponding to the analysis in Figure 4E.

To determine whether the observed connectivity patterns are specific to the dimensions of the micro-column or represent fundamental organizational principles of cortical circuitry, we performed a conceptual sub-sampling experiment restricted strictly to excitatory neurons in the micro-column (Fig. 5). By creating nested subnetworks representing fractions of the total volume—ranging from the full micro-column (1,139 E neurons) to progressively smaller scales of 0.5 (771 E neurons), 0.25 (398 E neurons), 0.125 (213 E neurons), and 0.0625 (122 E neurons)—we tested the stability of our above conclusions across varying spatial boundaries. We found that the core results of

Figures 2–4 remained remarkably consistent across all sub-volumes. We found that the number of local partners, both excitatory (Fig. 5B) and inhibitory (Fig. 5C) consistently increases with increase in spine density of the postsynaptic excitatory neuron. This distinct rising trend is present across all tested subnetwork scales, replicating the primary population-level results of Figure 2C. Similarly, when examining the preference for targeting spines as a function of the number of outgoing synapses (Fig. 5D), we observed that a positive correlation is generally maintained across the subnetworks. However, the strength of this correlation noticeably decreases as the subnetwork size shrinks. This attenuation is expected, as the progressive reduction in volume naturally leads to a smaller overall network and a corresponding drop in the number of outgoing synapses available for analysis.

Finally, we examined the relationship between shared input strength within the excitatory population across nested scales (Fig. 5E and Fig. S13). Consistent with our observations in the full network (Fig. 4E and Fig. S9), we found that neurons sharing more presynaptic inputs generally tend to exhibit a greater preference for targeting spines in their output connections. The full underlying data distributions for these subnetwork regressions are available in Figure S13.

Similar to Figure 5D, while the trends remain positive across the larger sub-volumes, the strength of the correlation gradually weakens with decreasing scale. Ultimately, at the smallest subnetwork tested (scaled by 0.0625), the correlations shown in Figures 5D–E lose statistical significance (Spearman $\rho = 0.02$, $p = \text{n.s.}$ and Pearson $R = 0.12$, $p = \text{n.s.}$, respectively). We attribute this decay in correlation strength to the sparse sampling within this minimal spatial volume, which is reduced to just 122 excitatory neurons and proportionally fewer synapses, rather than a fundamental break in the connectivity rule.

Discussion

In this study, we leveraged a large-scale volumetric MICrONS EM dataset to better understand the connectivity logic of dendritic spines in the mouse visual cortex. We begin by introducing an apparently counterintuitive result: sparsely spiny inhibitory neurons receive, on average, nearly twice as many incoming synapses, and at approximately twice the synaptic density, as spiny excitatory neurons (Fig. 1). This observation alters the traditional interpretation of dendritic spines, going back to Cajál (Ramón y Cajal, 1899), as devices whose primary function is to increase the number of synaptic contacts a neuron can receive (Yuste, 2011). Dense synaptic innervation clearly does not require spines as smooth dendrites can support abundant connectivity. This shifts the central question from whether spines enable synaptic inputs to what type of inputs or connectivity they enable. Here we propose that inhibitory interneurons and excitatory pyramidal neurons obey distinct connectivity strategies; inhibitory neurons are optimized for dense, direct innervation over their smooth dendritic shafts, whereas spiny pyramidal neurons may use spines to organize inputs according to higher-order connectivity principles, including ensemble formation.

In this study, we seek to uncover the structural role(s) of dendritic spines in cortical connectivity. We found that spine density robustly predicts both the absolute number of incoming synapses and the diversity of presynaptic partners (Fig. 2), providing direct structural validation for the “connectivity and diversity” hypothesis (Yuste, 2011). Yet the structural implications of spines went beyond expansion of the potential afferent connectivity (partners) of individual neurons. Rather, our analyses uncovered a highly organized network architecture driven by spines also on the efferent (output) side: for excitatory axons, the probability that an efferent synapse contacted a spine differed markedly between various neuron types and was strongly dependent on the axonal path distance of the (output) axonal synapse from the soma. The longer the axon, the stronger preference it has for spines (see also Schmidt et al. (2017) and Loomba et al. (2022)) and it emerged from highly divergent “broadcasting” neurons (Fig. 3). These differences were related to a neuron’s position within the network: neurons that become highly input-sharing precisely because they receive input from these broad projectors (see Appendix) also exhibit a strong tendency to target spines by their own outputs (Fig. 4). We term this “input-to-spine/output-to-spine organization”.

This structural motif provides a novel subcellular dimension for several foundational principles of cortical organization described in the literature. Foremost among these is the well-established principle of cortical clustering. The tendency for neurons sharing common inputs to form strongly interconnected functional groups (Perin et al., 2011) can thus be viewed not merely as a macroscopic network property, but as an architecture intrinsically linked to

spine-directed synaptic output. Other aspects of non-random connectivity structure have been linked to long-tailed degree distributions (Brunel, 2016), which in turn derive from lognormal distributions for axon lengths (Piazza et al., 2025). We have found that axon length and out-degree are also related to spines, specifically that highly divergent “broadcasting” neurons achieve their massive out-degrees through spine-directed connections (Fig. S5). Building on these principles, we also identified a novel, purely structural form of network assortativity. While assortativity - the tendency of similar nodes to connect - is a well-established functional feature of the cortex typically observed among neurons with similar stimulus tuning (Ko et al., 2011; Cossell et al., 2015), we found that it also exists at the subcellular targeting level. Specifically, excitatory neurons that receive input from divergent, spine-targeting axons tend to possess extended, spine-targeting axons themselves. Hence, wiring that is assortative with respect to subcellular targeting preference contributes to the generation of a tightly spine-based interconnected “rich club” (Van Den Heuvel and Sporns, 2011) of collaborative excitatory neurons.

Our findings also provide potential mechanistic explanations for previously described network phenomena. We speculate that the well-documented tendency for long and extended excitatory axons to target other excitatory neurons (Douglas and Martin, 2004; Petreanu et al., 2009; Yuste, 2011) might be merely a side effect of their higher likelihood to form spine synapses, combined with the higher spine density of excitatory neurons as compared to inhibitory neurons (Fig. 1). Rather than a complex wiring rule dictating specific cell-type connectivity, the structural expansion of highly divergent excitatory axons may be guided by this strict spine-targeting mechanism. Further supporting the concept that excitatory connectivity is driven by the search for spines rather than specific cell identities, we found that even among inhibitory post-synaptic neurons, those with higher spine densities attract a greater number of excitatory connections and a wider diversity of presynaptic partners (Fig. 2), suggesting an “excitatory-to-spines” rule, regardless of the post-synaptic cell type identity. Similarly, the previously described shift of target preference along excitatory axons from targeting inhibitory neurons with synapses near the soma to targeting excitatory neurons at higher distances from the soma (Schmidt et al., 2017; Bodor et al., 2025) may be explained by our observed shift in preferring spines vs. shaft as a function of the axon’s path distance from the soma (Fig. 3).

While we discuss “axonal spine-targeting preference”, the formation of a spine synapse happens on the postsynaptic side where spines grow out and meet pre-existing boutons (Yuste and Bonhoeffer, 2004; Nägerl et al., 2007; Zito et al., 2009; Yuste, 2023). We therefore should ask: what mechanism may explain the observed increase in spine targeting as the axon spreads away from the soma? We speculate that it may be a signal secreted from the axonal bouton, that is compared by nearby dendrites for similarity among themselves. If matching, growth of a spine towards the source bouton is triggered.

This hypothesis explains our results in the following ways. First, the proposed similarity matching leads to the observed assortativity. Second, the higher degree of spine targeting and dendritic spine density observed in some cell types (Fig. 4F) may indicate a higher similarity of that signal within that group. This would make it easier for individual axons of those types to find a matching dendrite that will then connect via a spine. Third, if we assume that continued axon growth is contingent on existing boutons finding a postsynaptic partner, then this also explains why axons with high spine-targeting preference are longer. This would mean that the previously described “rich club” of connectivity is a group of neurons whose homogeneity in signaling enables them to more easily find each other and connect. Fourth, the higher spine-targeting preference observed at axons with greater distances from the soma (Fig. 3) can be explained by an axonal signal responsible for inducing spine outgrowth becoming more potent (perhaps accumulation) in distal axon branches.

What are the functional consequences of distributing shared input through this distinct “spiny subnetwork”? Because the spines of different neurons preferentially share inputs originating from highly divergent hub cells, they create the necessary physical infrastructure for coordinated, synchronized network-wide activity. We therefore propose that dendritic spines constitute the structural backbone for network-level collaboration.

While subsequent population-level recordings have widely demonstrated the existence of coactive neuronal ensembles (Carrillo-Reid et al., 2016; Jennings et al., 2019), how this shared activity was physically organized across the circuit was unknown prior to the advent of large-scale connectomics. Leveraging the connectome, we are now able to directly measure the shared input between distinct neurons arising purely based on their connectivity.

We demonstrate that input sharing is inherently built into the cortical network via the connectivity statistics of spine-directed synapses, predominantly driven by “spine-preferring” populations (2/3P, 4P, and 5P-IT neurons, Fig. 4).

If we accept the premise that such collaboration is the structural prerequisite for the formation of neuronal ensembles and pattern completion, mathematically captured by attractor neural networks (Hopfield, 1982; Buzsáki, 2010), then it follows that these computational processes are fundamentally enabled at the dendritic compartment level (the spine) rather than the whole dendritic branch level (the shaft). This may explain why dendritic shafts, which we found receive less diverse, low-degree inputs (Fig. 3), are insufficient for covering the wide range of possible patterns required for cortical computation.

We conclude that evolution has produced two distinct classes of cortical neurons that operate according to different connectivity rules. Excitatory neurons are typically spiny, whereas inhibitory neurons tend to possess few, if any, dendritic spines. Decades of theoretical work (Rinzel and Rall, 1974; Segev and Rall, 1988; Koch and Zador, 1993; Segev and Rall, 1998), together with experimental studies using two-photon calcium imaging and, more recently, voltage imaging of individual spines, have shown that spines can function as partially isolated biochemical and electrical compartments (Yuste and Denk, 1995; Cornejo et al., 2022; Yuste, 2023) and can generate nonlinear, amplified voltage responses (Segev and Rall, 1988). Spines have also been shown to serve as specialized sites for synaptic plasticity (Yuste and Bonhoeffer, 2001; Yuste, 2011, 2023), thereby enabling cortical networks to adapt dynamically and store information.

Here, based purely on ultrastructural data, we provide structural support for the “connectivity and diversity” hypothesis, showing that dendritic spine density predicts both the absolute number of synaptic inputs and the diversity of presynaptic partners in both excitatory and inhibitory neurons. We also show that spines allow the excitatory network to achieve a higher degree of input cooperation, and at greater intensity, than the inhibitory population. This effect is particularly pronounced in the L2/3P, L4P, and L5P-IT subpopulations (Fig. 4). Importantly, the spine-based connectivity and cooperativity rules uncovered here operate across spatial scales, from the thinnest cortical “slices” containing only a few tens of neurons to the cubic-millimeter scale (tens of thousand neurons) covered by the MICrONS dataset (Fig. 5). An important next step will be to determine whether these connectivity rules continue to scale in larger cortical volumes. It will also be essential to examine, using computational simulations, how the “input-to-spine/output-to-spine” organization influences cortical network dynamics and learning.

Methods

MICrONS dataset annotation and structural tables

The MICrONS structural dataset consists of an electron microscopy volume of the visual cortex obtained from an adult mouse at postnatal day 87 (P87), corresponding primarily to the ‘minnie65’ reconstruction. All analyses utilized MICrONS dataset release version 1718 (March 7, 2026)¹. To establish the spatial locations and baseline anatomical identities of neurons within our micro-column, we relied on the `nucleus_detection_v0` and `allen_v1_column_types_slanted_ref` tables (Schneider-Mizell et al., 2025). Synaptic connectivity between these neurons was retrieved from the `synapses_pni_2` table using the CAVEclient `synapse_query` method.

To analyze the specific ultrastructural landing sites of individual connections across the network, we utilized the `synapse_target_predictions_ssa_v2` table (Pedigo et al., 2026). This table provided the compartmental predictions from which we derived the dendritic spine and dendritic shaft classifications used throughout the study.

Finally, to track manual curation and ensure topological accuracy for specific validations—such as the extended network analysis presented in Figure S3—we accessed the `proofreading_status_and_strategy` table². For this analysis, which included neurons spanning outside the boundaries of the micro-column, we strictly limited our analysis scope to a highly curated subset of neurons that had undergone exhaustive manual correction (referred to by the Allen Institute as “axon_fully_extended”). Because this extended analysis included neurons outside our

¹<https://tutorial.microns-explorer.org/annotation-tables.html>

²<https://tutorial.microns-explorer.org/proofreading.html>

primary volume, their structural classifications were based on automated predictions from the Allen Institute's `aibs_metamodel_celltypes_v661` table (Elabbady et al., 2025).

Neuron selection and dataset filtering

Our initial target population comprised 1,351 neurons (1,188 excitatory and 163 inhibitory) within the micro-column. While the original reference dataset (Schneider-Mizell et al., 2025) defines 1,352 neurons for this volume, a single skeleton was excluded due to an unrecoverable download failure via CAVE. To ensure we were working with fully resolved structural and synaptic data, we applied a few standard inclusion criteria to this initial cohort. First, we filtered out neurons with zero dendritic or axonal length, as well as any cells missing compartmental mapping for their incoming spine-directed synapses. We also calculated a synaptic tagging ratio for each cell—defined as the sum of synapses successfully mapped to the soma, dendritic shafts, and dendritic spines, divided by the total number of incoming synapses. We retained the neurons where this mapped synapse ratio exceeded 0.9. Following these basic filtering steps, our final analytical dataset consisted of 1,298 neurons (1,139 excitatory and 159 inhibitory).

Spine targeting distance along the axon

To quantify the spatial distribution of synaptic target preference, we analyzed the fraction of synapses targeting dendritic spines as a function of the axonal path distance from the presynaptic soma. To compute those distances from the presynaptic soma to individual synapses, pre-computed topological skeletal representations of the neurons were downloaded directly via the CAVEclient. Path lengths were then calculated for each synapse utilizing the built-in graph traversal logic within the MeshParty package (Dorkenwald et al., 2020). Outgoing synapses were sorted into discrete spatial bins (80–100 μm wide, depending on the specific figure). Within each bin, the spine-targeting fraction was computed as the number of synapses formed onto dendritic spines divided by the total number of outgoing synapses.

For population-level analyses (e.g., Fig. 3F, G, J, K, and Fig. S3), we utilized a neuron-weighted (“macro”) approach. The average spine fraction per bin was calculated as the mean of the individual axonal fractions within that bin, with population variance represented by the standard error of the mean (SEM). A synapse-weighted (“micro”) approach, which instead pools all synapses across the population prior to calculating the fraction, yielded highly similar results.

To precisely identify the transition from a shaft-biased to a spine-dominated targeting regime (Fig. S7), we determined a characteristic crossing distance—the path length where exactly 50% of synapses target spines. This threshold was mathematically derived via linear interpolation between the centers of the two consecutive bins bracketing the 0.5 fractional crossing: $x_0 + (x_1 - x_0) \cdot [(0.5 - y_0)/(y_1 - y_0)]$, where (x_0, y_0) and (x_1, y_1) represent the distance and spine fraction of the bins immediately before and after the threshold, respectively.

At the individual neuron level, crossing distances exhibited substantial diversity and a heavily right-skewed distribution. To capture this biological variance, we categorized single-cell behaviors into three distinct groups: (1) continuously spine-preferring neurons, which maintained a spine fraction above 0.5 from their first valid proximal bin (assigned a crossing distance of 0 μm); (2) transitioning neurons, which exhibited a clear crossing point within the reconstructed volume; and (3) shaft-preferring neurons that never crossed the 0.5 threshold along their traced length (assigned a crossing distance of ∞). Because the population-level curve (which averages the fractions per bin) naturally smooths over this heterogeneity, single-cell transition points (e.g., Fig. S7) are summarized using the median and interquartile range (IQR) of the finite crossing values, alongside the specific population percentages of the 0 μm and ∞ boundary cases.

Statistical analysis

To rigorously test the statistical significance of the shared input properties presented in Figure 4B, we employed a “blocked” configuration (CFG) null model. This degree-preserving null model was constructed by separately shuffling each of the four connectivity blocks (Excitatory-to-Excitatory (EE), Excitatory-to-Inhibitory (EI), Inhibitory-to-Excitatory (IE), and Inhibitory-to-Inhibitory (II)) within the adjacency matrix. By preserving the in- and out-degree

distributions for each neuron within these specific sub-circuits, this method established a more constrained and robust baseline expectation compared to shuffling the entire network as a single piece.

To evaluate the multimodality of shared input strength distributions (Fig. 4B), we applied Hartigan's Dip Test (Hartigan and Hartigan, 1985). This analysis was implemented using the diptest Python package³, which calculates the Dip statistic and associated *p*-value to test the null hypothesis of unimodality.

Neuron identification in figures

The specific neurons visualized in the representative figures are identified using their `pt_root_ids` from the MICrONS dataset release version 1718. Reconstructions presented in Figures 1 and 2 display only dendritic morphology, whereas the neurons in Figure 3 show both dendrites and axons. The selected neurons, along with their corresponding color coding used in the visual renderings, are detailed as follows:

- **Figure 1B:** Layer 2/3 pyramidal neuron (red), `pt_root_id` 864691135277186789; Layer 5 basket interneuron (blue), `pt_root_id` 864691135875777166.
- **Figure 2A:** Layer 4 pyramidal neuron (purple), `pt_root_id` 864691136618553563; Layer 6 IT pyramidal neuron (green), `pt_root_id` 864691135214643256.
- **Figure S1:** Layer 2/3 Martinotti interneuron (purple), `pt_root_id` 864691135571356038; Layer 5 basket interneuron (green), `pt_root_id` 864691135875414414.
- **Figure 3A:** Layer 5 IT pyramidal neuron (purple), `pt_root_id` 864691135617152361.
- **Figure 3B:** Layer 6 CT pyramidal neuron (green), `pt_root_id` 864691136177632518.
- **Figure 3C:** Layer 4 basket interneuron (blue), `pt_root_id` 864691135562001633.

Spine density trend analysis and binning

To robustly analyze the relationship between dendritic spine density and incoming connectivity metrics, mitigating the influence of extreme biological outliers, we grouped the neurons into specific density bins for the analyses presented in Figure 2 (and Fig. S2).

In our raw data, the measured spine density ranged from 0.1 to 1.8 spines/ μm for excitatory neurons (Fig. 2C) and 0.04 to 1.2 spines/ μm for inhibitory neurons (Fig. S1C). To create a consistent scale for our trendlines across these figures, we divided the data into custom bins with centers ranging from 0.25 to 1.25 spines/ μm (using bin edges at 0.125, 0.375, 0.625, 0.875, 1.125, and 1.375). We then calculated both the average connectivity metric and the standard error of the mean (SEM) for the neurons within each bin. This approach provided reliable central tendencies to plot the data and allowed us to calculate the overall changes and multiplication factors across the analyzed trendline.

Code availability

All code produced for this study will be made available upon publication.

Acknowledgements

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³Software implementation provided by R. Urlus; <https://github.com/RUrlus/diptest>

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Appendix. Normalized Common Incoming Neighbors and Incoming-Source Out-Degree Are Order-Equivalent

This document outlines the structural equivalence between the normalized sum of common incoming neighbors and the incoming-source out-degree in directed graphs.

Setup

Let $G = (V, E)$ be a directed finite graph with $n = |V|$ nodes (where $n > 1$). Define the binary adjacency matrix $M \in \{0, 1\}^{n \times n}$ where $M_{wu} = 1$ if and only if there is a directed edge from w to u .

For a node $v \in V$, let:

- $N^-(v) = \{w \in V : w \rightarrow v\}$ be the in-neighborhood of v .
- $d^-(v) = |N^-(v)|$ be the in-degree of v .
- $d^+(w)$ be the out-degree of a node w .

Raw Quantities

Define two unnormalized quantities for each node v :

Definition 1 (Total Common Incoming Neighbors). Summed over all other nodes:

$$\tilde{A}(v) = \sum_{u \neq v} |N^-(u) \cap N^-(v)| \quad (1)$$

Definition 2 (Total Out-Degree of Incoming Sources).

$$\tilde{B}(v) = \sum_{w \in N^-(v)} d^+(w) \quad (2)$$

Key Lemma

Lemma 1. For any node $v \in V$,

$$\tilde{A}(v) = \tilde{B}(v) - d^-(v) \quad (3)$$

Proof. Express the intersection count using the adjacency matrix M :

$$\tilde{A}(v) = \sum_{u \neq v} \sum_{w \in V} M_{wu} M_{wv}$$

Swap the order of summation:

$$\tilde{A}(v) = \sum_{w \in V} M_{wv} \sum_{u \neq v} M_{wu}$$

The inner sum counts the out-edges of w , excluding the edge to v . This holds true whether or not the graph contains self-loops:

$$\sum_{u \neq v} M_{wu} = d^+(w) - M_{wv}$$

Substituting this back into the equation:

$$\tilde{A}(v) = \sum_{w \in V} M_{wv} d^+(w) - \sum_{w \in V} M_{wv}^2$$

Because M is a binary matrix, $M_{wv}^2 = M_{wv}$. Therefore, the second sum is exactly the in-degree $d^-(v)$, yielding:

$$\tilde{A}(v) = \tilde{B}(v) - d^-(v)$$

□

Normalized Metrics and Their Relationship

For any node v where $d^-(v) > 0$, we define the normalized metrics:

$$A(v) = \frac{\tilde{A}(v)}{(n-1)d^-(v)} \tag{4}$$

$$B(v) = \frac{\tilde{B}(v)}{d^-(v)} \tag{5}$$

Dividing both sides of Lemma 1 by $(n-1)d^-(v)$ yields:

$$\frac{\tilde{A}(v)}{(n-1)d^-(v)} = \frac{\tilde{B}(v)}{(n-1)d^-(v)} - \frac{1}{n-1}$$

Which simplifies to the exact identity:

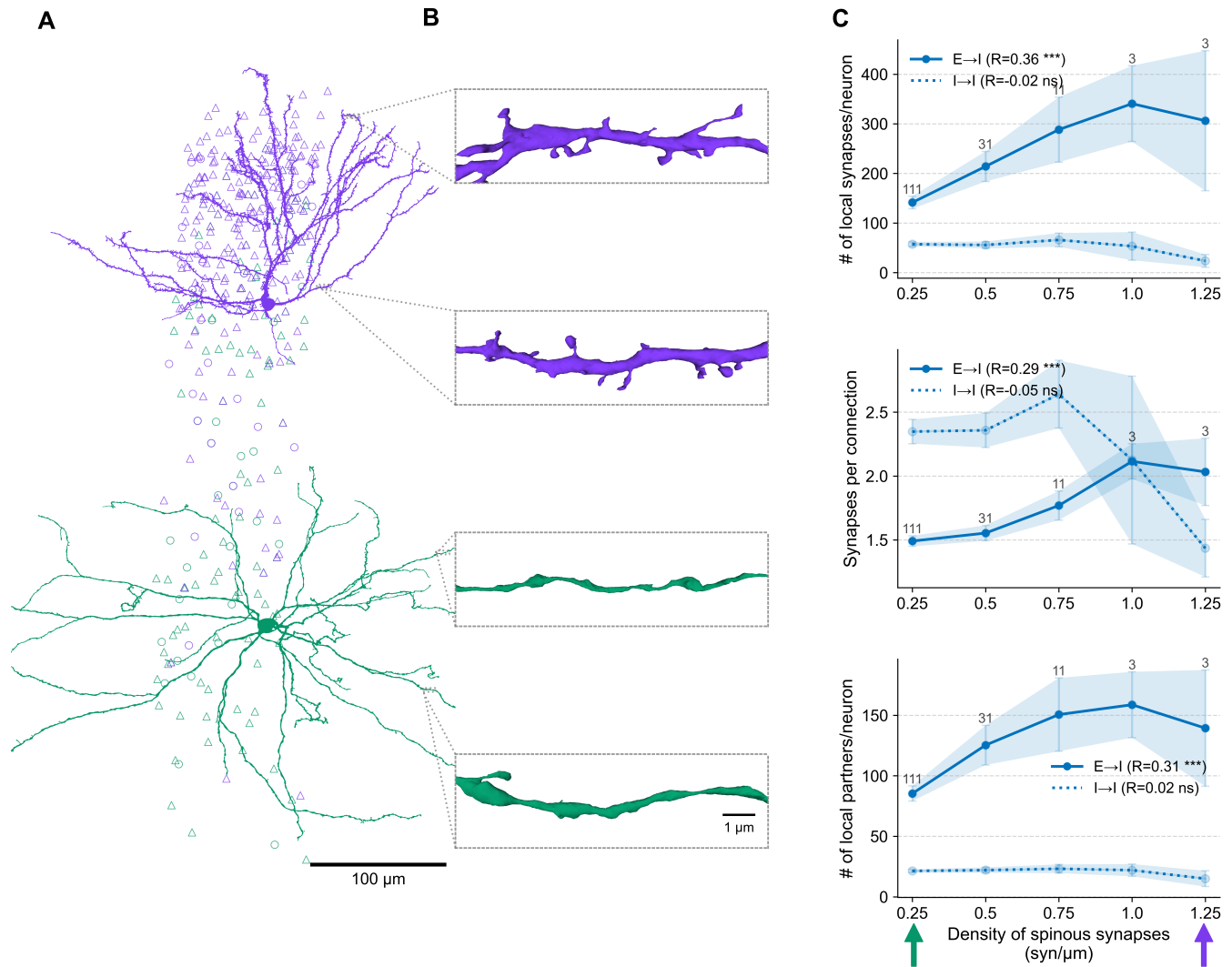
$$A(v) = \frac{B(v) - 1}{n-1} \tag{6}$$

Order Equivalence

Corollary 1. *Because the transformation $A(v) = \frac{B(v)-1}{n-1}$ is strictly increasing with respect to $B(v)$, the following holds for any nodes $u, v \in V$ (provided both have an in-degree greater than zero):*

$$A(u) \geq A(v) \iff B(u) \geq B(v) \tag{7}$$

Supplementary Figures



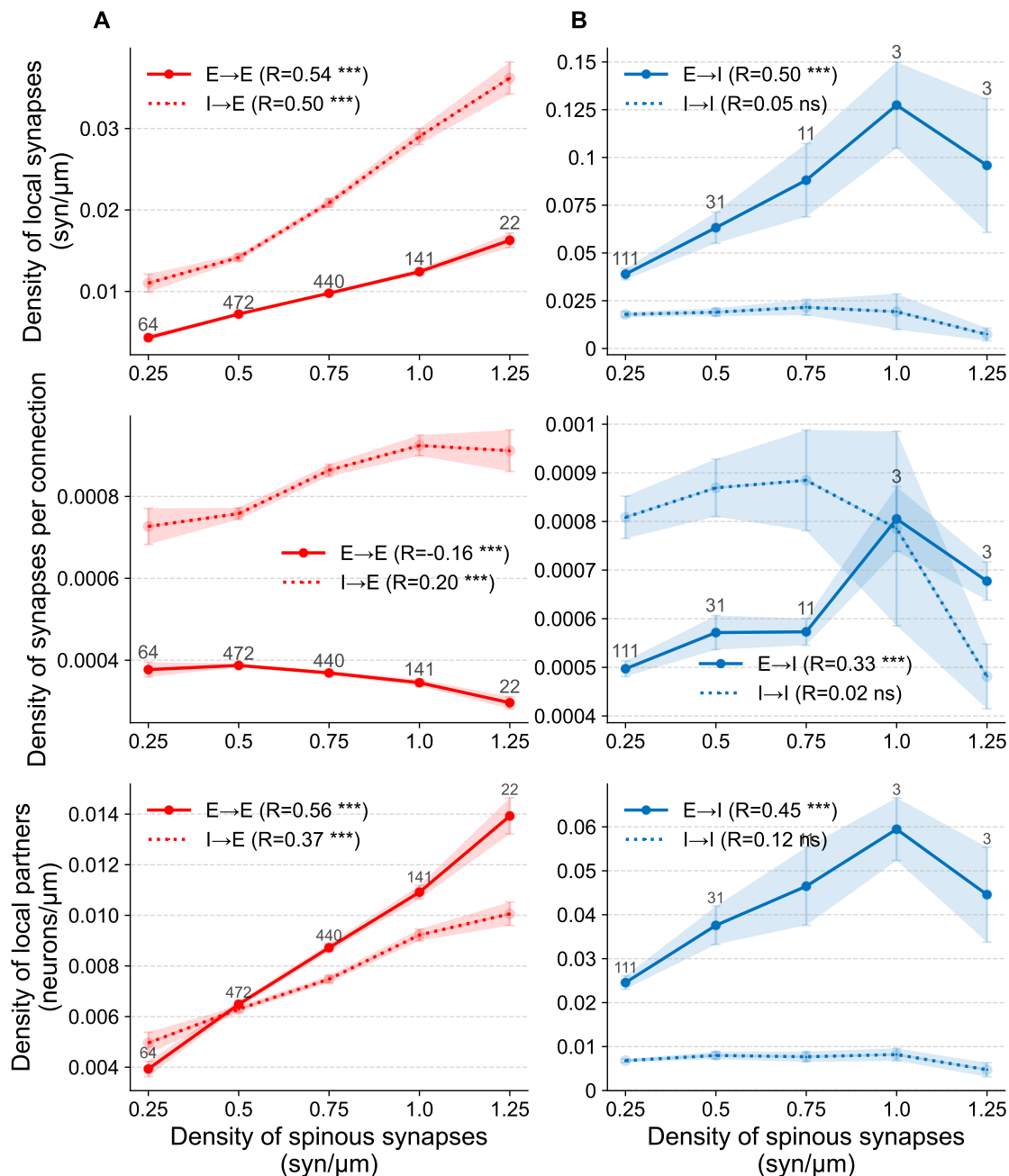


Figure S2. Synaptic input richness and diversity normalized by dendrite length. **A.** Normalized metrics versus total spine density for excitatory neurons. Unlike Figure 2C, all y-axis values are normalized by the total dendrite length of the (post-synaptic) excitatory neuron. *Top:* Density of local synaptic inputs as a function of spine density. *Middle:* Density of synapses per connection as a function of spine density. *Bottom:* Density of distinct presynaptic partners as a function of spine density. Solid lines: excitatory-to-excitatory synapses; dashed: inhibitory-to-excitatory synapses. **B.** As in A, but for inhibitory neurons (corresponding to Fig. S1C).

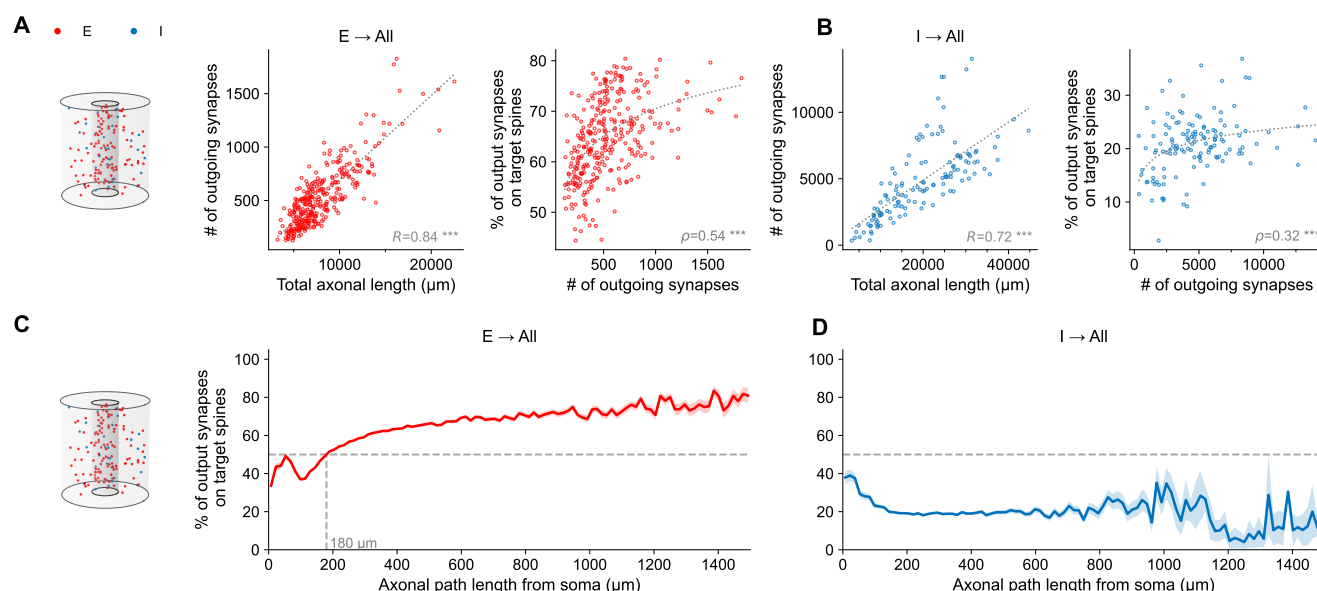


Figure S3. Distance-dependent spine targeting is robust across the most carefully proof-read reconstructed axons. Analyses as in Figure 3 but on a strict subset of neurons with the highest level of manual proofreading of their axons, consisting of 272 excitatory and 7 inhibitory neurons from within the micro-column, plus 42 excitatory and 126 inhibitory neurons from outside the micro-column. **A.** Analysis of synapses made by excitatory axons originating both inside and outside the micro-column onto the whole reconstructed circuit (schematic at left). Neurons outside the micro-column are classified based on the Allen Institute *aibs_metamodel_celltypes_v661* table (Elabbady et al., 2025) (and see Methods). *Left:* Number of excitatory axonal synapses correlates linearly with total axonal length. *Right:* Percentage of excitatory synapses onto dendritic spines increase as a function of the total number of outgoing synapses also in this highly curated group of axons. **B.** As in A but for the inhibitory axons. *Left:* Number of inhibitory synapses is linearly correlated with axonal length. *Right:* Percentage of synapses onto spines shows weaker dependence on the number of outgoing synapses. **C.** Spine targeting as a function of path distance along excitatory axons. The monotonic increase and spatial transition from a mixed regime near the soma to a spine-dominated regime at larger distances remain clearly evident. **D.** Same as in C for inhibitory axons. Spine targeting remains low and largely independent of path distance along the inhibitory axon.

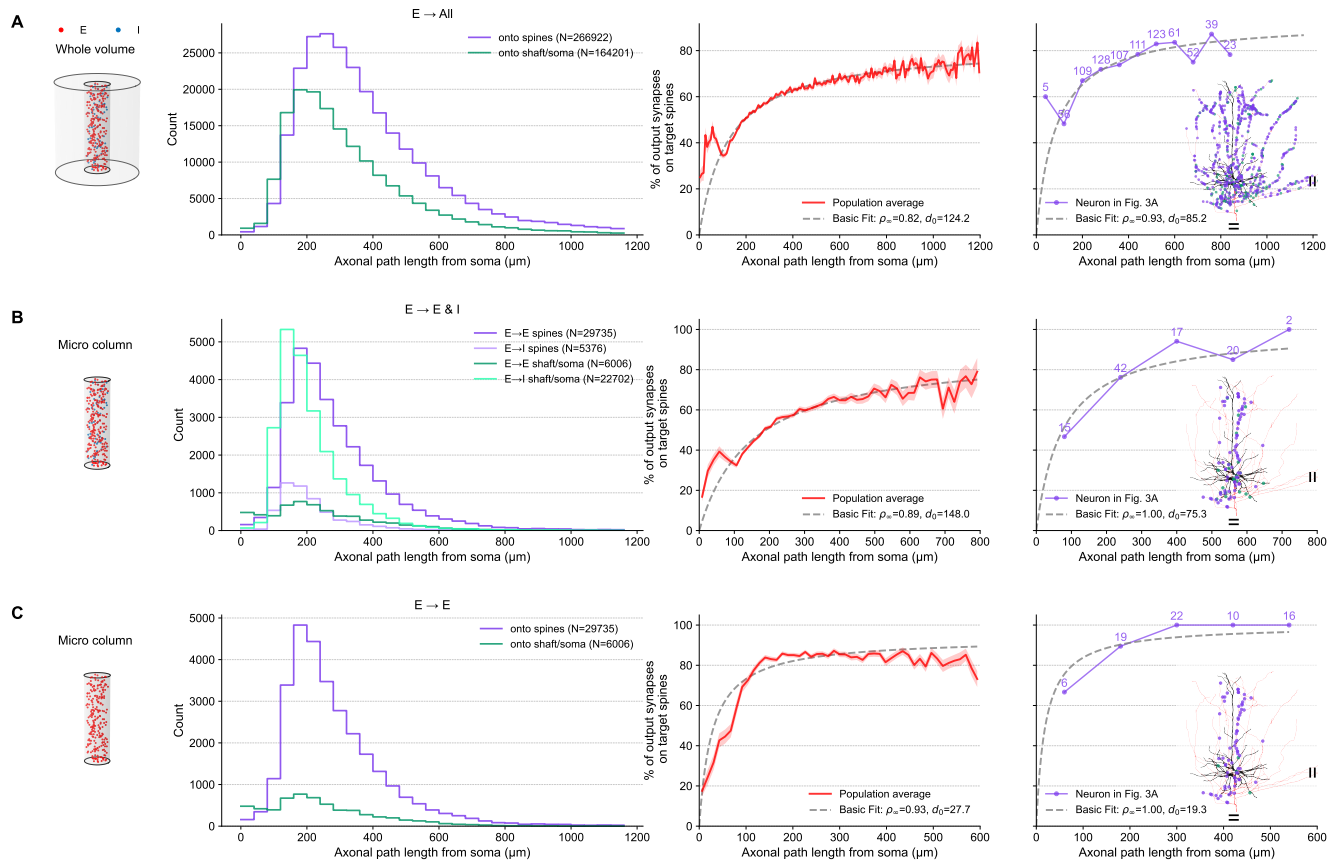


Figure S4. A minimal model of spine targeting preference as a function of axonal path length. **A.** Whole-volume analysis of all outgoing excitatory synapses. *Left:* Distribution of the number of synapses along the axonal path length, categorized by their target (purple for spines, green for shafts). *Middle:* Population-level target preference showing the percentage of synapses targeting spines (in red). The fitted model using Eq. (1) is shown by the dashed line; $\rho_{\infty} = 0.82$, $d_0 = 124$. *Right:* As in Middle but for a single representative L5-IT pyramidal neuron, also shown in Fig. 3A (number of synapses per bin are indicated above); fitted model is shown by the dashed line ($\rho_{\infty} = 0.93$, $d_0 = 85.2$). **B.** Analysis of local excitatory connections restricted to the micro-column. *Left:* Synapse distributions for four local targeting categories: onto spines of excitatory (E→E) and inhibitory (E→I) neurons (dark and light purple, respectively), and onto shafts or somas of E and I neurons (dark and light green, respectively). *Middle:* Population-level model fit for all targets within the local micro-column ($\rho_{\infty} = 0.89$, $d_0 = 148$). *Right:* Single-neuron fit restricted to these local micro-column targets ($\rho_{\infty} = 1.00$, $d_0 = 75.3$). **C.** Local E→E subnetwork analysis within the micro-column. *Left:* Distribution of synapses restricted exclusively to excitatory-to-excitatory connections, targeting spines (purple) versus shafts/somas (green). *Middle:* Population-level model fit for the E→E subnetwork ($\rho_{\infty} = 0.93$, $d_0 = 27.7$). *Right:* Single-neuron model fit for local E→E targets ($\rho_{\infty} = 1.00$, $d_0 = 19.3$).

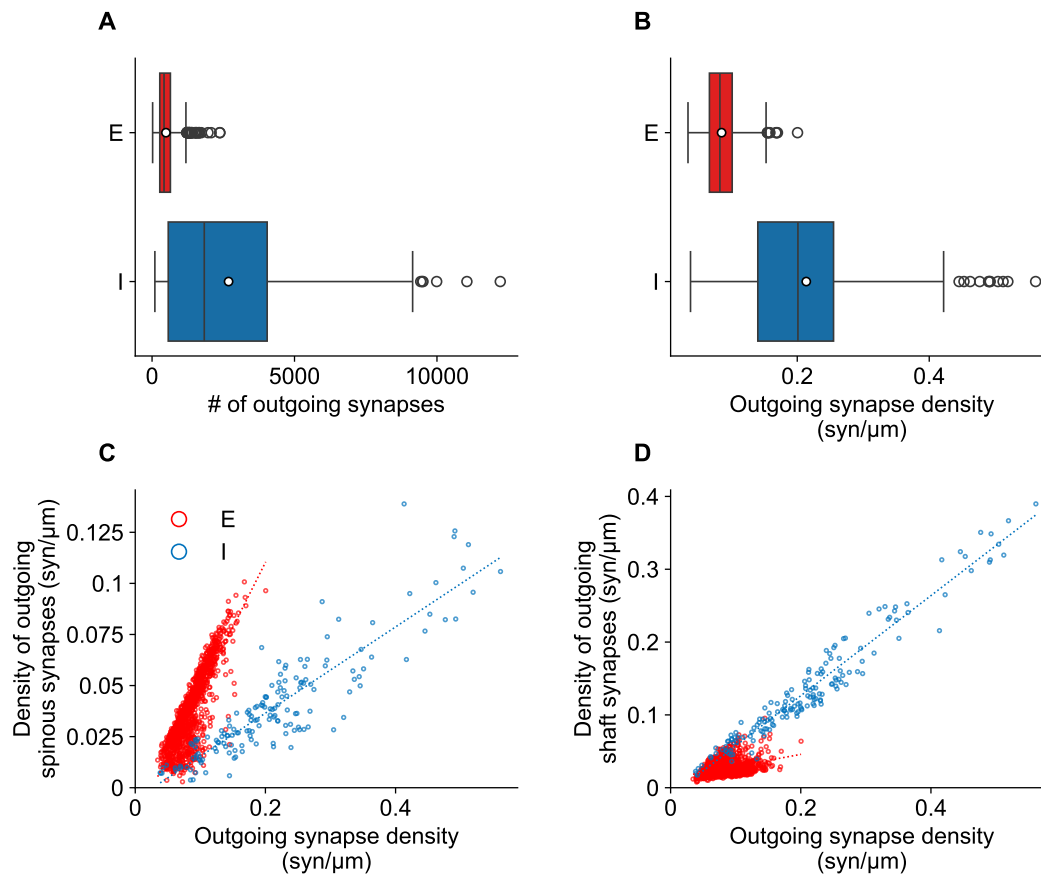


Figure S5. Divergent axonal targeting strategies: Excitatory subtypes exhibit diverse target preferences, while inhibitory neurons uniformly target dendritic shafts. **A–B.** Distribution of outgoing synaptic outputs for excitatory (red) and inhibitory (blue) neurons. **A.** The total number of outgoing synapses per neuron; **B.** Density of outgoing synapses. In both panels, the box edges represent the 25th and 75th percentiles, the vertical line denotes the median, and the white circle indicates the mean. Whiskers extend to 1.5 times the interquartile range. Mean $\pm$ SD values: **A,** excitatory: 488 ± 319 vs. inhibitory: $2,686 \pm 2,623$; **B,** excitatory: 0.09 ± 0.02 vs. inhibitory: 0.21 ± 0.11 . **C.** Density of outgoing spinous synapses as a function of total outgoing synapse density for excitatory and inhibitory neurons. Linear regressions are shown by the dotted lines (excitatory: slope = 0.21, $R^2 = 0.80$, $p < 0.001$; inhibitory: slope = 0.17, $R^2 = 0.16$, $p < 0.001$). As a population, excitatory neurons show a robust correlation between total number of output synapses and number of spine targeted. **D.** Density of outgoing shaft synapses as a function of total outgoing density. Linear regressions are shown by dotted lines (excitatory: slope = 0.63, $R^2 = 0.71$, $p < 0.001$; inhibitory: slope = 0.69, $R^2 = 0.95$, $p < 0.001$). The scaling of inhibitory output density is driven almost entirely by shaft-targeting synapses.

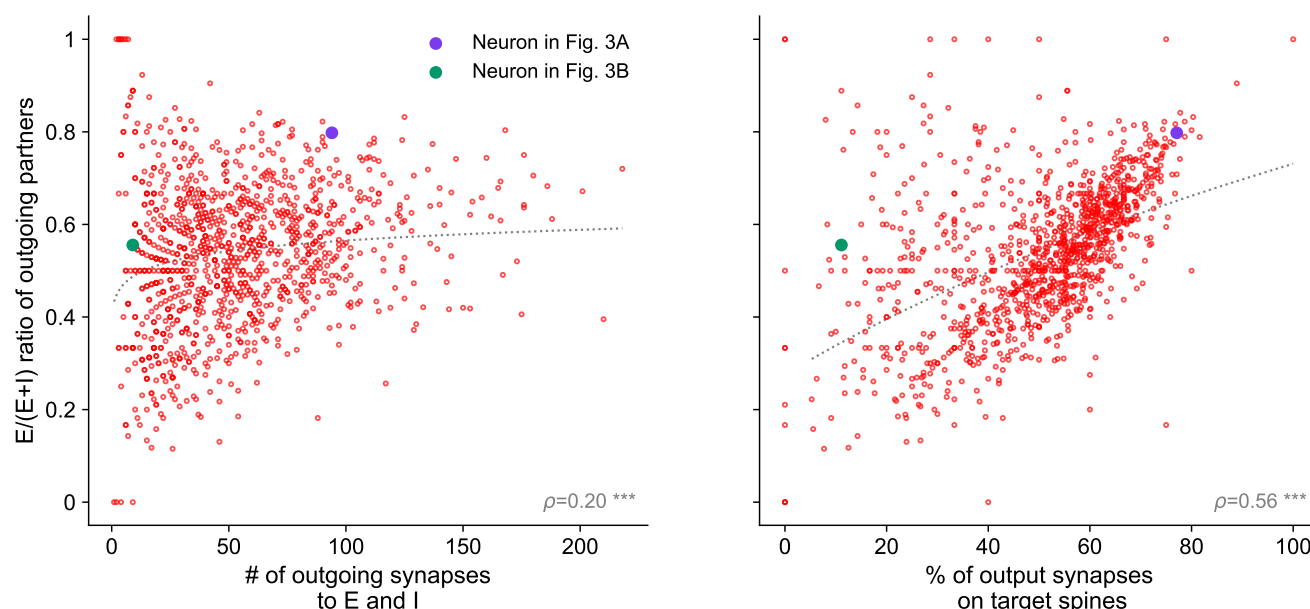


Figure S6. Spine preference is a stronger predictor of excitatory partner selection than total synaptic output. Analysis of local connectivity restricted to the micro-column, evaluating both excitatory and inhibitory targets for presynaptic excitatory neurons (red circles). The y-axis in both panels represents the target cell-type preference, calculated as the ratio of unique excitatory partner neurons to total local partner neurons $[E/(E+I)]$. *Left:* The ratio of excitatory partners shows only a weak correlation with the total number of outgoing synapses to both E and I targets (Spearman $\rho = 0.20$, $p < 0.001$). *Right:* The neuron's overall preference for targeting dendritic spines (the percentage of output synapses onto local spines) is a much stronger predictor for its preference to connect with excitatory partners (Spearman $\rho = 0.56$, $p < 0.001$). Highlighted neurons correspond to the representative L5-IT (purple, neuron in Figure 3A) and L6-CT (green, neuron in Figure 3B) cells shown in Figure 3. This indicates that the tendency to target excitatory neurons is tightly coupled to an underlying subcellular preference for dendritic spines.

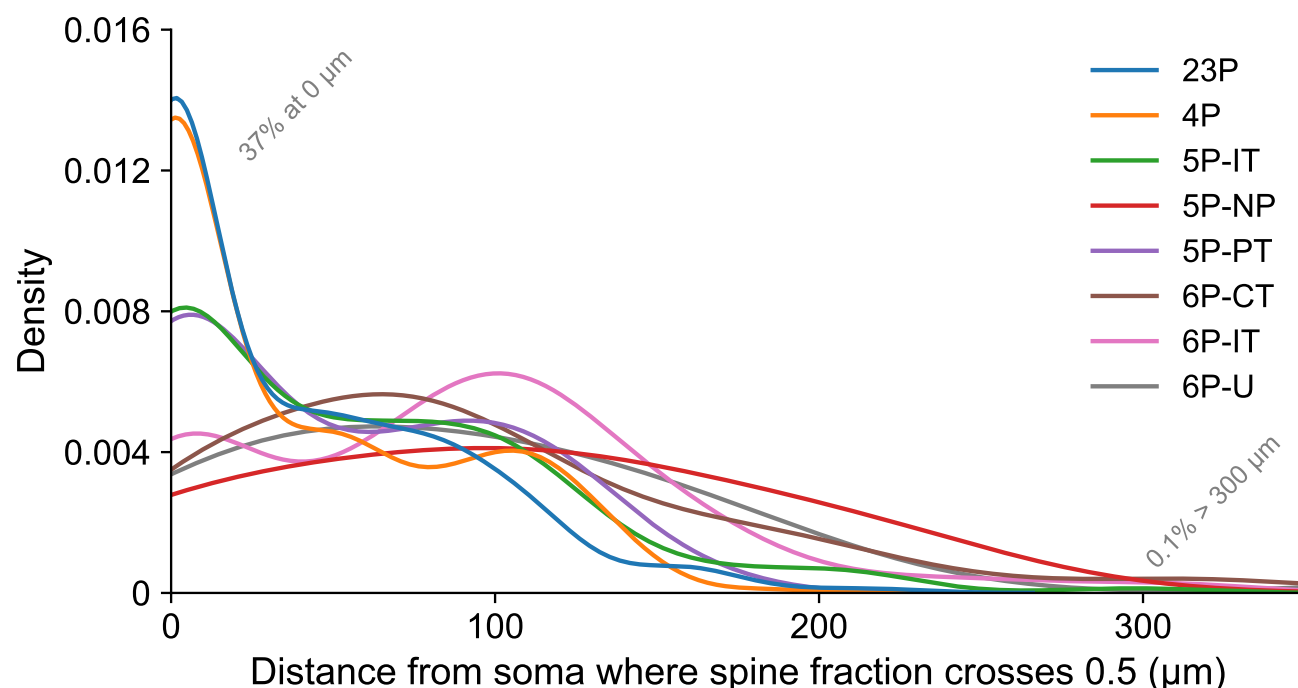


Figure S7. Distribution of axonal path distance from soma where spine targeting fraction crosses 50%. Kernel density estimate (KDE) of the distance from the soma where the spine-targeting fraction crosses the 50% threshold for individual excitatory neurons. This analysis incorporates all outgoing synapses from the E population within the 1 mm^3 volume, serving as an extension to the population-level analysis in Figure 3F. Distributions are color-coded by cell type, with each normalized independently. Note that a substantial subset of neurons (37%) are "always above" the threshold (i.e., their crossing distance is 0 μm), preferring spines throughout, while a negligible fraction (0.1%) never reaches the 50% threshold within the volume, i.e., remained shaft-preferring across their entire reconstructed length.

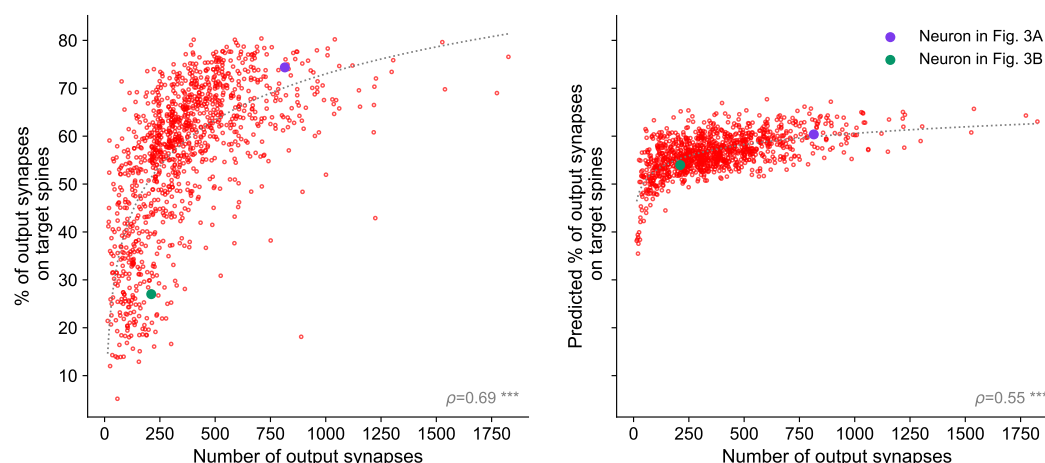


Figure S8. Spine preference variability cannot be fully explained by distance-based axonal mass distribution. Comparison of observed versus predicted spine preference for individual excitatory neurons. *Left:* Observed global spine preference as a function of total outgoing synapses per neuron (as in Figure 3D). *Right:* Predicted spine preference for the same neurons. Predictions rely solely on spatial distribution, generated by passing the empirical path length of every outgoing synapse through the population-fitted minimal model ($\rho_{\infty} = 0.82$, $d_0 = 124.2 \mu\text{m}$). While this distance-based model preserves a baseline upward trend (Spearman $\rho = 0.55$, $p < 0.001$), it fails to reproduce the wide empirical variance and the full magnitude of spine preference seen on the left.

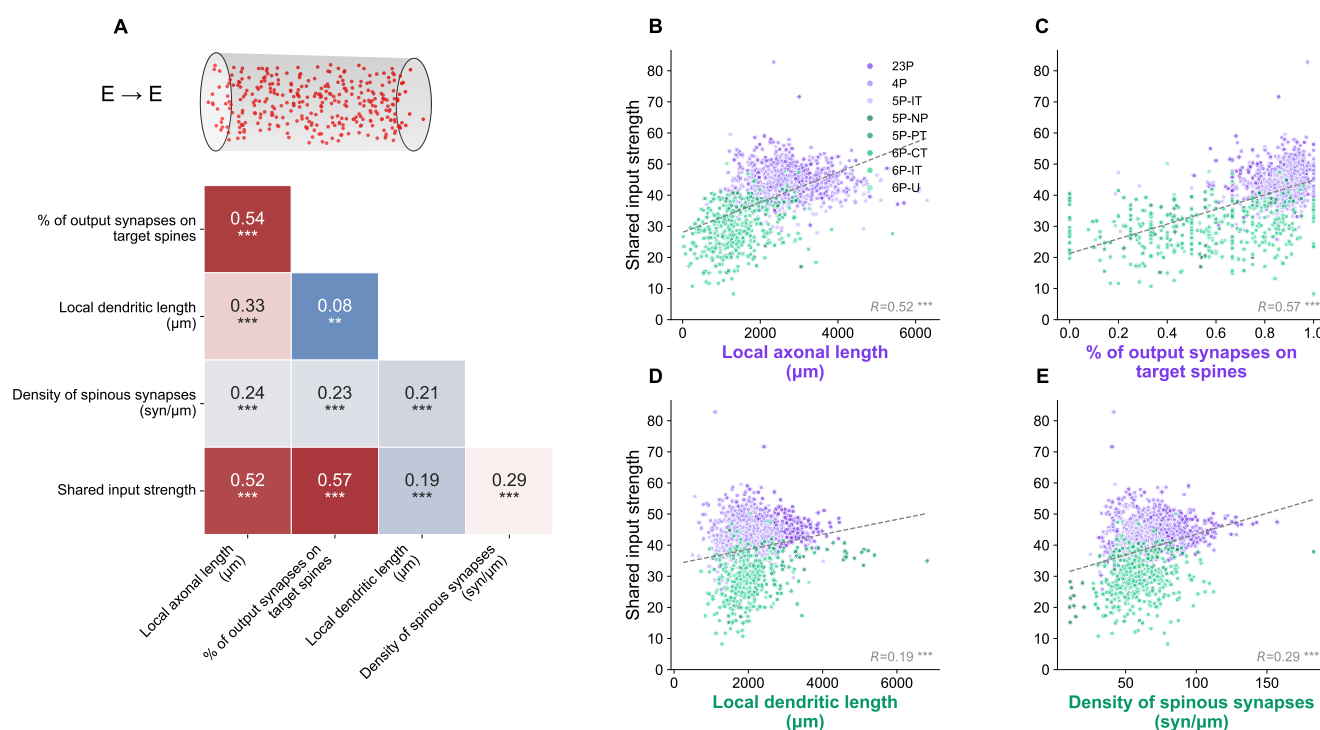


Figure S9. Feature correlations determining shared input strength within the excitatory sub-network (E-E connectivity). *A. Top:* Schematic representation indicating that this analysis is restricted to the excitatory-to-excitatory (E-E) connections within the micro-column. *Bottom:* A 5×5 Pearson correlation matrix displaying the relationships between shared input strength and four distinct structural features. For clarity, only the lower triangular portion of the matrix is shown, omitting the mirrored upper half and the diagonal of ones. *B-E:* Scatter plots detailing the relationship between a neuron's shared input strength (y-axis) and individual network features (x-axis) for the E-E sub-network. Neurons are color-coded based on their cell type, with high-sharing neurons (2/3P, 4P, 5P-IT) in purple and the remaining neurons in green (classification detailed in main text). Panels B and C display correlations with “output” features (Pearson $R = 0.52$, $p < 0.001$ and $R = 0.57$, $p < 0.001$, respectively), which notably exhibit stronger correlations, while panels D and E display correlations with “input” features (Pearson $R = 0.19$, $p < 0.001$ and $R = 0.29$, $p < 0.001$, respectively) with the shared input strength.

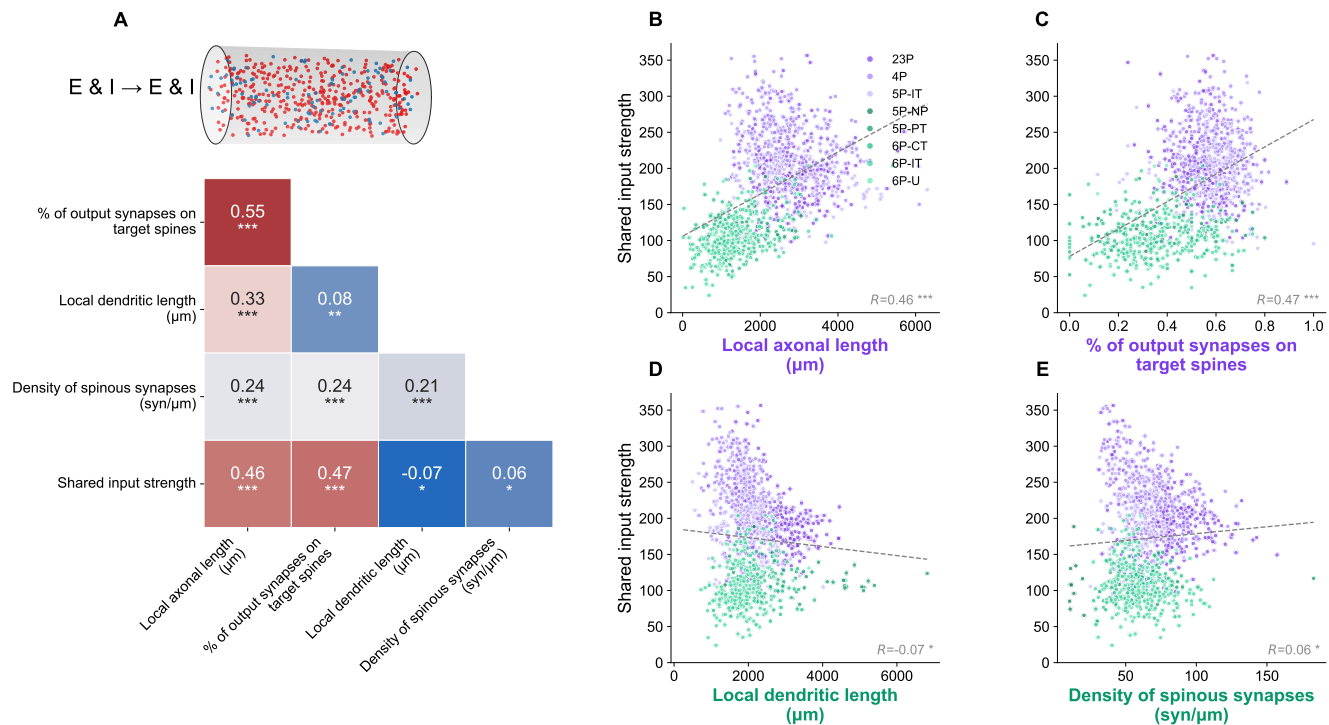


Figure S10. Feature correlations determining shared input strength across the full micro-column network (combined E and I inputs). This figure follows the exact same analysis and layout as Fig. S9, but extends the evaluation to the full network, incorporating both excitatory and inhibitory inputs to each neuron (matching the analysis conditions of main Fig. 4). **A:** A 5×5 lower-triangular Pearson correlation matrix showing the relationships between shared input strength and the four structural features for the full network. **B–E:** Full scatter plots of shared input strength (y-axis) versus the four individual features (x-axis), with high-common-input neurons indicated in purple and the rest in green. Similar to the isolated E-E network, “output” features B–C show stronger correlations (Pearson $R = 0.46$, $p < 0.001$ and $R = 0.47$, $p < 0.001$, respectively) with shared input strength than “input” features D–E (Pearson $R = -0.07$, $p = 0.019$ and $R = 0.06$, $p = 0.045$, respectively).

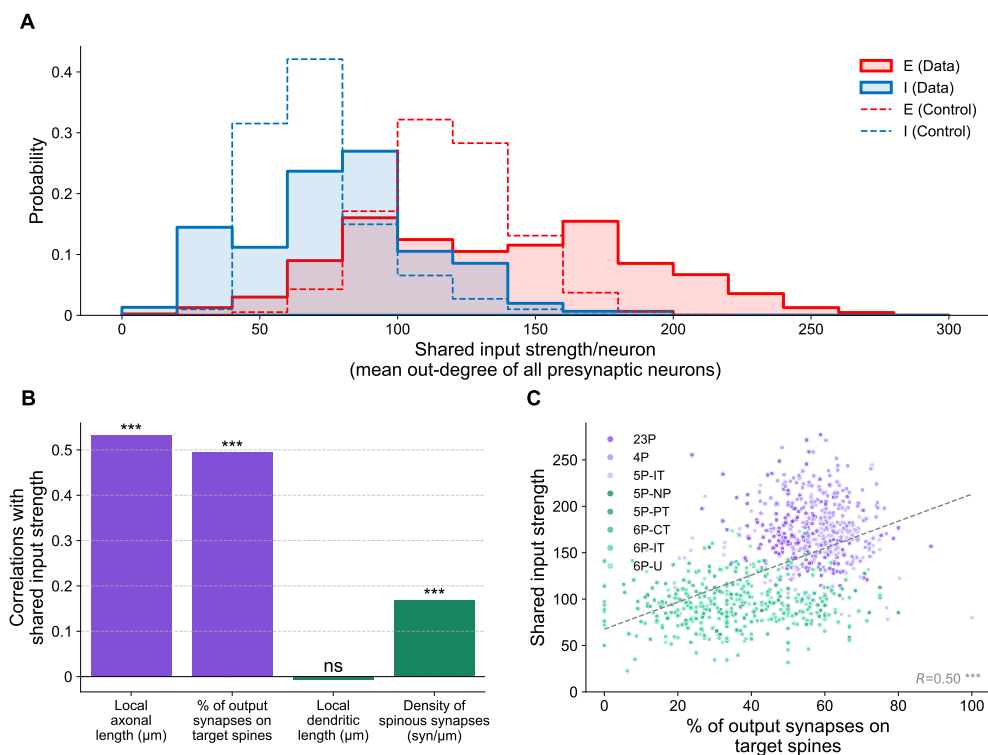


Figure S11. Shared-input architecture and preferential spine targeting excluding fully reconstructed axons. Analysis equivalent to Figure 4, performed on the “axon_partially_extended” neuronal subset (excluding 272 fully proofread excitatory micro-column neurons), where the input to each neuron consists of both excitatory and inhibitory connections. **A**, Distribution of the shared input strength per neuron for the excitatory (solid red) and inhibitory (solid blue) populations, compared against null distributions from 100 degree-preserving “blocked” configuration (CFG) models (dashed lines). **B**, Correlation of the shared input strength against four anatomical and connectivity features (purple: output features; green: input features). **C**, Scatter plot detailing the relationship between a neuron’s shared input strength and its spine preference (percentage of output synapses on target spines) across the entire micro-column population.

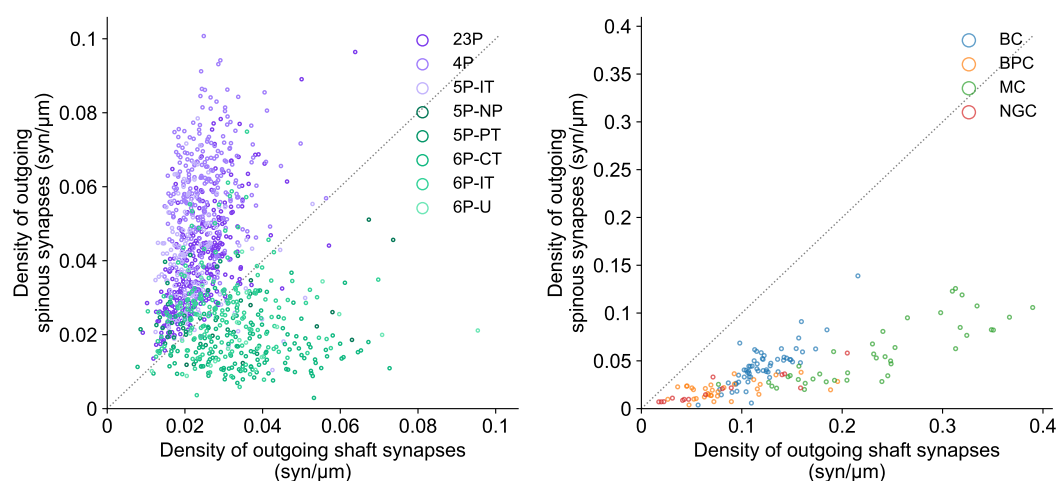


Figure S12. Subtype-specific axonal targeting strategies: Excitatory populations exhibit diverse target preferences, while inhibitory neurons uniformly target dendritic shafts. This is a whole-volume analysis evaluating the ~1M outgoing synapses that project outside the micro-column. Direct comparison of outgoing spinous versus shaft synapse density for individual excitatory (Left) and inhibitory (Right) neurons, colored by cell subtype. The dotted grey line represents the 1:1 identity line (equal density on spines and shafts). *Left*: Excitatory neurons display broad, subtype-specific diversity in their targeting strategies. The high-sharing subpopulations (purple) predominantly target spines (clustering above the unity line), whereas the deeper-layer and output populations (green) exhibit a stronger preference for dendritic shafts (clustering below the unity line). *Right*: In contrast, all identified inhibitory subtypes exhibit a universal strategy, clustering entirely below the identity line to exclusively target dendritic shafts regardless of subtype classification.

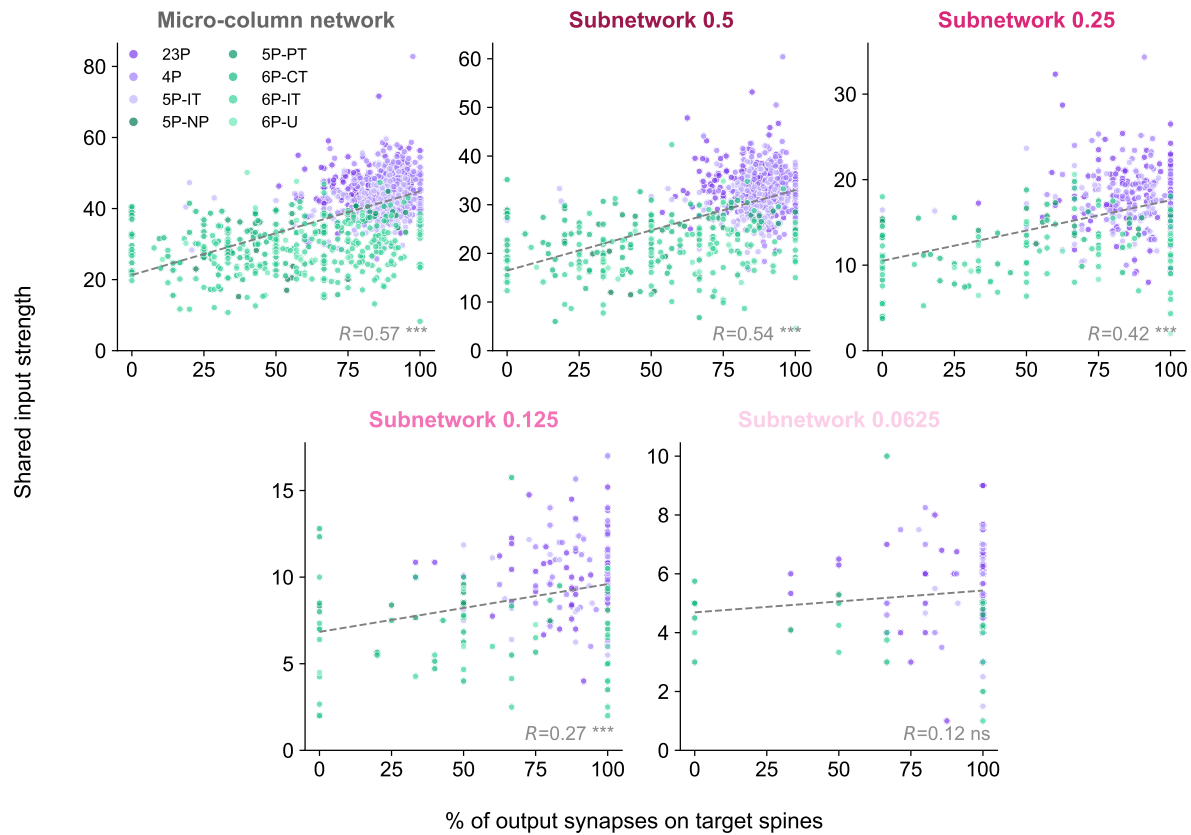


Figure S13. Correlation between shared input strength and percentage of output synapses on target spines across subnetwork scales. Corresponding to the analysis in Figure 5E. Neurons are colored by cell type: high-sharing, spine-preferring populations (purple) and deeper-layer, shaft-preferring populations (green). The positive correlations are maintained in larger volumes, but visible dispersion increases and correlation decays in the smallest subnetworks due to sparse sampling.